

# **Safety Briefing**

**Soh Eng Keng**

**Safety Committee Chairperson**

**Engineering Design and Innovation Centre**

## **2 Important tasks to do after the Safety Briefing:**

- 1) Safety Quiz**
- 2) Risk Assessment**

# Introduction

As you start your hands on work for your project, you are likely to use the CDE/EDIC facilities and equipment for your work.

For your health and safety, there are some facility and safety requirement which you need to comply with.

You must also be aware of some general safety guidelines to ensure your own and others' safety.

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- 1) Introduction to the EDIC Safety Committee
- 2) Introduction to the Facility ICs and Facilities
- 3) Safety Videos
- 4) Risk Assessment / Safe Work Procedures
- 5) Mandatory Eye Protection
- 6) General Safety Rules / Expectation
- 7) Covered / Closed-toes shoes policy
- 8) Storage Facilities
- 9) Emergency Services and Contacts
- 10) Fire Evacuation and Emergency Assembly
- 11) Workshop/Lab User Registration
- 12) Safety Awareness Quiz

# Introduction to the EDIC Safety Committee

|                          |                                                                |
|--------------------------|----------------------------------------------------------------|
| Advisor:                 | Dr. Yen Shih-Cheng                                             |
| Chairperson:             | Mr. Soh Eng Keng                                               |
| Vice Chairperson:        | Mr. Neo Kang Wei, Kenneth                                      |
| First Aider (E2A):       | Mr. Ee Wei Han, Eugene                                         |
| Block E2A Admin Manager: | Mr. Alvin Poh                                                  |
| E2A Fire Safety Team:    | Ms. Tan Annie (Electronics Workshop/E2A Fire Warden Level 2)   |
|                          | Ms. Woo Mei Yin (Electronics Workshop/E2A Fire Warden Level 3) |
|                          | Ms. Jillian Teo (E2A Fire Warden Level 4)                      |
| Member:                  | Mr. Lim Hong Wee (FSAE Lab)                                    |

# Introduction to the Facility ICs

| Workshops/Labs                      | Location                                                                                                                                 | Workshop Supervisor | Laboratory Technologist/Executive                                            |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------|
| iDP Electronics Workshop            | E2A-02-02                                                                                                                                | A/Prof Lim Li Hong  | Ms. Annie Tan                                                                |
| iDP Assembly Workshop 1             | E1A-01-13                                                                                                                                | Mr. Royston Shieh   | Mr. Patrick Tan                                                              |
| iDP Assembly Workshop 2             | E1A-01-04                                                                                                                                | Mr. Royston Shieh   | Mr. Patrick Tan                                                              |
| EDIC EIM Laboratory (Gait Lab)      | E1A-01-05                                                                                                                                | Mr. Soh Eng Keng    | Mr. Azmall Fraiszudeen (BME) – Lab Executive                                 |
| CDE Central Workshop                | E1A-01-06 (Training Workshop)<br>E3-01-06 (Professional Workshop)<br>E3-B1-06 (Professional Workshop)<br>E4-02-06 (Fabrication Workshop) | Mr. Hamilton Toh    | Mr. Foo Check Kuang<br>Ms. Pearlyn Neo<br>Mr. Dickson Foo<br>Mr. Vincent Bay |
| FSAE Workshop                       | E4-B1-03                                                                                                                                 | Mr. Lim Hong Wee    | Ms. Woo Mei Yin                                                              |
| Innovation and Design Hub (the Hub) | EA-04-04 to EA-04-07                                                                                                                     | A/Prof Lim Li Hong  | Ms. How Yee Teng                                                             |

# Design Studios



iDP Studio 1 / E2A



iDP Studio 5 / E2A



iDP Studio 2 / E2A

**NO POWERED TOOLS ALLOWED**



iDP Studio 3 / E2A



iDP Studio 6 / EA



iDP Studio 4 / E2A

# EDIC Facilities



# CDE Facilities

Fabrication Workshop



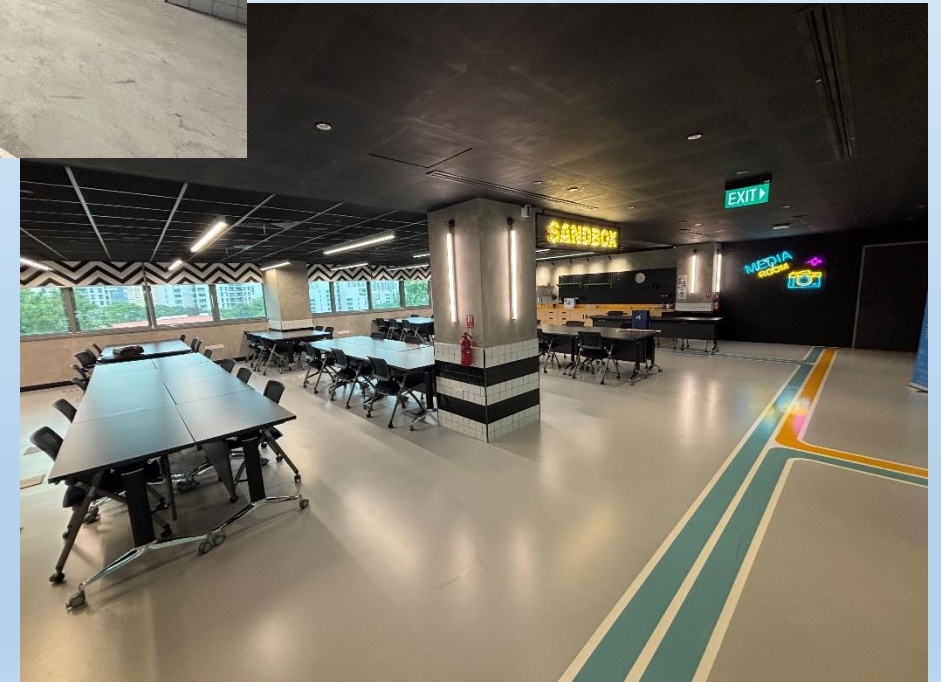
Professional Workshop



Training Workshop



## Innovation and Design Hub (The Hub)



# **What you need to do in order to gain access to iDP workshops and Labs**

# Safety Videos

- Introduction to Laboratory Safety and Health in NUS  
(to view after class)
- Dept of Mechanical Engineering Safety Video  
(to view after class)
- First Aid Video (Singapore Emergency Responder Academy)  
(to view after class)

Uploaded to NUS Canvas (Safe845 - Files – EDIC Safety Awareness Quiz – Safety Videos)

<https://canvas.nus.edu.sg/courses/43200/files/folder/Safety%20Videos>

# What you need to do in order to gain access to iDP workshops and Labs

## A) To gain access to Workshops and Labs

- 1) Approach Workshop/Lab LT/Supervisor
- 2) Read and understand safety guidelines
- 3) Be briefed on emergency response plan
- 4) Watch Safety Videos / attempt and **pass safety awareness quiz**
- 5) Fill up and submit Workshop/Lab user registration form to LT
- 6) Create/submit **Project/Activity Risk Assessment (RA)** for access to and work in Workshops/Labs
- 7) Create/submit Safe Work Procedure (SWP) if necessary

## B) To use Workshop and Lab equipment

- 1) Satisfy all the steps in Part A)
- 2) Read and understand the Risk Assessment (RA) and Safe Work Procedure (SWP) of relevant tools, equipment and activities
- 3) **Be trained** by respective LTs to use the respective equipment
- 4) Follow requirements specific to each different workshop/lab – check with LT in charge.

To clear any doubts, approach and consult any member of the Safety Committee

# Recap

- 1) Pass the Safety Awareness Quiz
- 2) Conduct Risk Assessment for your project before you begin hands on work.

## **Risk Assessment (RA)**

- 1) RA is a document that allows you to assess the hazards and risks that exist in your project, and put in the appropriate controls to eliminate or minimize these risks.
- 2) RA is a requirement for organizations under the Workplace Safety Act.
- 3) RA is not a one time process. You may need to review and update it when there are changes along the way

## **When do you conduct your RA?**

You need to conduct your RA before you begin your hands on work. But your project nature may change, so you need to constantly review your RA and do the necessary updates.

# What can be exempted from inclusion in an RA

## Pure Simulation Work

- Writing Computer Program
- Computer Aided Design and Analysis

Note: once you plug in a wire to a circuit board to test your program, it is no longer pure simulation work.

# Risk Assessment (RA)

A requirement under the law

Two types:

- 1) Project based RA
- 2) Activity or Experiment-based RA

You may download the blank forms for generating the RA from NUS Canvas:

Safe845 - Files – EDIC Safety Awareness Quiz – Risk Assessment Templates

<https://canvas.nus.edu.sg/courses/43200/files/folder/Risk%20Assessment%20Templates>

Or EDIC e-station – Module – Safety - Risk Assessment forms

[https://canvas.nus.edu.sg/courses/43124/pages/risk-assessment-forms?module\\_item\\_id=133976](https://canvas.nus.edu.sg/courses/43124/pages/risk-assessment-forms?module_item_id=133976)

# Project based RA

When you are doing a project, it will most likely involve more than one activity or experiment.

The project based risk evaluation allows you to think through all the activities in your project and assess their risks.

- 1) For each activity, you will have to check if a standard RA and SWP are available. If they are, you have to carefully read and understand them, and include them in your Project RA.
- 2) If your activity is not found in the standard RA/SWP archive, for e.g., you are conducting a new experiment, then you will have to create an activity/experiment based RA and attach/reference it in your project RA.

## Example

As an example of what is required to be stated in the Project RA form, if your project task requires the use of the laser cutting machine, then you need to read and understand the Activity RA and SWP of the laser cutting machine, and be trained by the LT in charge before using it.

A list of standard RA and SWP for the standard tools and equipment in EDIC, with the respective document numbers, has been posted onto iDP e-Station (Canvas) – Safety folder.

For the above example, the document numbers (to state) of the RA and SWP for the laser cutting machine are FAB/RA/01 and FAB/SOP/01 respectively.

# Project RA

# Project Based Risk Evaluation Form

(download from iDP e-station or Safe845)



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Project Risk Evaluation

**Note:** Project Supervisors are to discuss with their project students and fill up the following information. In the event that activities conducted have risks which are not listed by the Risk Assessment (RA) and Safe Work Procedures (SWP) provided by the Safety Committee, Project Supervisors and students are to identify all of such risks and identify risk controls. Project Supervisors and students must submit this completed form to the Safety Committee before embarking on the activities.

Reference No:

It is the responsibility and duty of all personnel (including students) in the project to read and understand all RA and SWP related to the required tools/equipment/machinery before completing this form. By submitting this form, it is deemed that all personnel assumed full responsibility and liability.

| Project Details                                                    |            |                              |                             |  |
|--------------------------------------------------------------------|------------|------------------------------|-----------------------------|--|
| Project Title                                                      |            |                              |                             |  |
| Project Supervisor                                                 |            |                              |                             |  |
| Project Duration                                                   | Start Date |                              | End Date                    |  |
| Team members                                                       |            |                              |                             |  |
| Location where Project will be carried out                         |            | Location 1                   |                             |  |
|                                                                    |            | Location 2                   |                             |  |
|                                                                    |            | Location 3                   |                             |  |
| Were the specific Workshop Supervisors / Lab Technicians Informed? |            | <input type="checkbox"/> Yes | <input type="checkbox"/> No |  |

| Experiment / Activity Steps (please list the information you have at this moment to the best of your knowledge) |          |                       |                        |
|-----------------------------------------------------------------------------------------------------------------|----------|-----------------------|------------------------|
| No.                                                                                                             | Location | Experiment / Activity | Workable Sequence/Task |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |
|                                                                                                                 |          |                       |                        |

1: fill up document number

2: fill up project details

3: fill up the activities which you/your group will be involved in

# Project Based Risk Evaluation Form

Workshops / Labs

| Workshops / Labs you might need to access |                     |          |                                                 |
|-------------------------------------------|---------------------|----------|-------------------------------------------------|
| S/No.                                     | Workshop / Lab name | Location | Workshop / Lab Equipment standard RA available? |
|                                           |                     |          |                                                 |
|                                           |                     |          |                                                 |
|                                           |                     |          |                                                 |
|                                           |                     |          |                                                 |
|                                           |                     |          |                                                 |

Equipments

| Electric Hazards                                                                                                                                                                                                |                     |                                                             |                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of Electrical Appliance                                                                                                                                                                                    | Current/Voltage use | Hazard Description                                          | Risk Control Implemented                                                                                                                                                                                                             |
| Please state devices only under the following cases:<br>1) the device you is not listed in the tools/machines registry from EDIC<br>2) the use of tool/machine deviates from the standard SWP provided by EDIC. |                     | (Please describe the kind of hazard which might be present) | <ul style="list-style-type: none"><li>Residual Current/ earth Leakage devices used</li><li>Equipment grounded to earth</li><li>Equipment is tested/tag</li><li>Others – to specify (e.g., created SWP/RA document numbers)</li></ul> |
|                                                                                                                                                                                                                 |                     |                                                             |                                                                                                                                                                                                                                      |
|                                                                                                                                                                                                                 |                     |                                                             |                                                                                                                                                                                                                                      |
|                                                                                                                                                                                                                 |                     |                                                             |                                                                                                                                                                                                                                      |

| Mechanical/ Physical Hazards                                                                                                                                                                                    |                                                                                                                                                                                                                |                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of Equipment/Tool                                                                                                                                                                                          | Hazard Description                                                                                                                                                                                             | Risk Control Implemented                                                                                                                                                                                                |
| Please state devices only under the following cases:<br>1) the device you is not listed in the tools/machines registry from EDIC<br>2) the use of tool/machine deviates from the standard SWP provided by EDIC. | Please describe the kind of hazard which might be present: <ul style="list-style-type: none"><li>Cutting</li><li>Crushing</li><li>Striking</li><li>Holist/Lifting Device</li><li>Others – to specify</li></ul> | <ul style="list-style-type: none"><li>Machine Guarding</li><li>Safety Interlocks</li><li>Safety Stop buttons</li><li>Lock out, tag out procedures during ser</li><li>Others – to specify (e.g., created SWP/)</li></ul> |
|                                                                                                                                                                                                                 |                                                                                                                                                                                                                |                                                                                                                                                                                                                         |
|                                                                                                                                                                                                                 |                                                                                                                                                                                                                |                                                                                                                                                                                                                         |
|                                                                                                                                                                                                                 |                                                                                                                                                                                                                |                                                                                                                                                                                                                         |

1: fill up the locations and all possible hazards that might be present in the activities and how you can mitigate them. If there is none in any category, state NA.

Chemical / Biological Hazards

| Type of Chemical/ Biological substances                                                                                                                                  | Hazard Description                                          | Risk Control Implemented               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------|
| (Please state clearly the type of chemical/ Biological substances to be used and how it is to be used during the project, including the handling, storage and disposal.) | (Please describe the kind of hazard which might be present) | (or <u>created</u> RA document number) |
|                                                                                                                                                                          |                                                             |                                        |
|                                                                                                                                                                          |                                                             |                                        |
|                                                                                                                                                                          |                                                             |                                        |

Activities / Experiments

| Will there be an experiment / test / project activity carried out? |                            |                                                                                               | Yes / No |
|--------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------|----------|
| S/No.                                                              | Activity / Experiment name | Activity / Experiment Risk Assessment required (state RA number or planned date to create RA) |          |
|                                                                    |                            |                                                                                               |          |
|                                                                    |                            |                                                                                               |          |
|                                                                    |                            |                                                                                               |          |
|                                                                    |                            |                                                                                               |          |

2: list any other hazards which do not belong to the above categories and the risk controls

| Other Hazards                                                                                                                                                                   |                                                             |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------|
| Type of Device/Tool/Machine                                                                                                                                                     | Hazard Description                                          | Risk Control Implemented |
| (If the device you use is not listed in the tools/machines registry from EDIC or the use of tool/machine deviates from the standard SWP provided by EDIC, please state it here) | (Please describe the kind of hazard which might be present) |                          |
|                                                                                                                                                                                 |                                                             |                          |
|                                                                                                                                                                                 |                                                             |                          |
|                                                                                                                                                                                 |                                                             |                          |

# Project Based Risk Evaluation Form

Summarise the main hazard

State SDS validity for chemicals

Declaration

| Risk Summary             |                           |
|--------------------------|---------------------------|
| Identify the main Hazard | Describe the main control |
|                          |                           |
|                          |                           |

**Declaration** (Please tick one)

- ☐ Risks posed is mild/moderate and unlikely to increase during the course of project
- ☐ Risks are significant but will be effectively controlled
- ☐ There is uncertainty about the level of risks
- ☐ Risks are significant and cannot be effectively controlled

- \* For Chemical Hazards, please ensure that the **SDS** (Safety Data Sheet) is attached.
- ☐ SDS Attachment (please tick if they are attached with this form)  
SDS Validity Period: \_\_\_\_\_ to \_\_\_\_\_ (please indicate the dates)

|                                      |  |  |  |  |  |
|--------------------------------------|--|--|--|--|--|
| Conducted by:<br>(Name, designation) |  |  |  |  |  |
| Signature:                           |  |  |  |  |  |
| Date:                                |  |  |  |  |  |

|                                  |  |
|----------------------------------|--|
| Approved by: (Name, designation) |  |
| Signature:                       |  |
| Date:                            |  |

# Illustration (Project Risk Evaluation)



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Project Risk Evaluation

**Note:** Project Supervisors are to discuss with their project students and fill up the following information. In the event that activities conducted have risks which are not listed by the Risk Assessment (RA) and Safe Work Procedures (SWP) provided by the Safety Committee, Project Supervisors and students are to list all of such risks and identify risk controls. Project Supervisors and students must submit this completed form to the Safety Committee before embarking on the activities.

Reference No: RA/CDE3301/IS399/2025

It is the responsibility and duty of all personnel (including students) in the project to read and understand all RA and SWP related to the required tools/equipment/machinery before completing this form. By submitting this form, it is deemed that all personnel assumed full responsibility and liability.

| Project Details                                                    |                                                 |                                                          |                          |             |
|--------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------|--------------------------|-------------|
| Project Title                                                      | Assembly and Testing of a Window Cleaning Robot |                                                          |                          |             |
| Project Supervisor                                                 | Thomas Jobs                                     |                                                          |                          |             |
| Project Duration                                                   | Start Date                                      | 14 Feb 2025                                              | End Date                 | 04 Dec 2025 |
| Team members                                                       | Tan Siew Peng                                   | Mohammad Ali                                             | Raveen Sharma            |             |
|                                                                    | Tan.siewpeng@u.nus.edu                          | Mod.Ali@u.nus.edu                                        | Sharma.raveen@u.nus.edu  |             |
| Location where Project will be carried out                         |                                                 | Location 1                                               | iDP Electronics Workshop |             |
|                                                                    |                                                 | Location 2                                               | iDP Assembly Workshop 1  |             |
|                                                                    |                                                 | Location 3                                               | iDP Studio 1             |             |
| Were the specific Workshop Supervisors / Lab Technicians Informed? |                                                 | <input type="checkbox"/> Yes <input type="checkbox"/> No |                          |             |

| Experiment / Activity Steps (please list the information you have at this moment to the best of your knowledge) |                          |                                |                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------|
| No.                                                                                                             | Location                 | Experiment / Activity          | Workable Sequence/Task                                                                                                 |
| 1                                                                                                               | IDP Electronics Workshop | Assembly & Fitting of tool kit | Drilling holes on Trolley<br>Attach 3D printed tool kit and fastening to the holes                                     |
|                                                                                                                 |                          | Assembly of Robot Arm          | Attach linear actuator to frame on trolley<br>Attach robotic arm to linear actuator<br>Connect Electronics and battery |
|                                                                                                                 |                          | Charging of Battery            | Connect battery to charger or power socket                                                                             |

Note: all names used here are for illustration purpose only. No identification with actual persons is intended or should be inferred.

# Illustration (Project Risk Evaluation)

|   |                         |                                                           |                                                                                                                                                                                                                                 |
|---|-------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | iDP Assembly Workshop 1 | Assembly & Testing of pneumatic system                    | Connect tubes for compressed air tank, solenoid valve and pneumatic tool changer.<br>Connect wiring for solenoid valve to microcontroller<br>Charge up air tank<br>Testing of system                                            |
|   |                         | Assembly & Testing of Water Pump System                   | Connect battery and electronics<br>Connect hose from water pump to robotic arm<br>Testing of System                                                                                                                             |
| 3 | iDP Studio 1            | Placing battery, CPU and other Electronics inside trolley | Connect wires from battery/perf board to the respective components<br>Testing of system                                                                                                                                         |
|   |                         | Testing of Robot after assembly                           | Pre-checks<br>Starting Program from LCD screen<br>Pushing trolley to window till image detected<br>Cleaning process by robotic arm<br>Testing Emergency button<br>Shutting down program & moving trolley back to charging point |
|   |                         |                                                           |                                                                                                                                                                                                                                 |
|   |                         |                                                           |                                                                                                                                                                                                                                 |
|   |                         |                                                           |                                                                                                                                                                                                                                 |
|   |                         |                                                           |                                                                                                                                                                                                                                 |

# Illustration (Project Risk Evaluation)

## Workshops / Labs

| Workshops / Labs you might need to access |                                |                         |                                                 |
|-------------------------------------------|--------------------------------|-------------------------|-------------------------------------------------|
| S/No.                                     | Workshop / Lab name            | Location                | Workshop / Lab Equipment standard RA available? |
| 1                                         | iDP Electronics Workshop       | E2A-02-02               | Yes                                             |
| 2                                         | iDP Assembly Workshop 1 & 2    | E1A-01-04 and E1A-01-13 | Yes                                             |
| 3                                         | iDP Studio 1-6                 | E2A                     | No (no equipments in the studios)               |
| 4                                         | CDE Central Workshop (Fab Lab) | E4-02-06                | Yes                                             |
|                                           |                                |                         |                                                 |
|                                           |                                |                         |                                                 |

## Equipments

| Electric Hazards                                                                                                                                                                                                                                |                     |                                                                                   |                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of Electrical Appliance<br>Please state devices only under the following cases:<br>1) the device you is not listed in the tools/machines registry from EDIC<br>2) the use of tool/machine deviates from the standard SWP provided by EDIC. | Current/Voltage use | Hazard Description<br>(Please describe the kind of hazard which might be present) | Risk Control Implemented<br><ul style="list-style-type: none"><li>Residual Current/ earth Leakage devices used</li><li>Equipment grounded to earth</li><li>Equipment is tested/tag</li><li>Others – to specify (e.g., created SWP/RA document numbers)</li></ul> |
| Water pump                                                                                                                                                                                                                                      | 12V                 | Electrical Shock                                                                  | To create Activity RA                                                                                                                                                                                                                                            |
| Battery Charger                                                                                                                                                                                                                                 | 12V                 | Short circuit<br>Overcharging<br>Sparks                                           | Please take reference from EDIC/RA/EW/080 Standard RA                                                                                                                                                                                                            |
| Testing of battery and other electronics after placing in trolley                                                                                                                                                                               | 12-24V              | Sparks<br>Short circuit                                                           | To create Activity RA                                                                                                                                                                                                                                            |
| Pneumatic system setup (air compressor)                                                                                                                                                                                                         | 12V                 | Pressurized air<br>Short circuit                                                  | To create Activity RA                                                                                                                                                                                                                                            |

# Illustration (Project Risk Evaluation)

| Mechanical/ Physical Hazards                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of Equipment/Tool<br><small>Please state devices only under the following cases:</small><br>1) the device you is not listed in the tools/machines registry from EDIC<br>2) the use of tool/machine deviates from the standard SWP provided by EDIC. | Hazard Description<br><small>Please describe the kind of hazard which might be present:</small> <ul style="list-style-type: none"><li>Cutting</li><li>Crushing</li><li>Striking</li><li>Hoist/Lifting Device</li><li>Others – to specify</li></ul> | Risk Control Implemented <ul style="list-style-type: none"><li>Machine Guarding</li><li>Safety Interlocks</li><li>Safety Stop buttons</li><li>Lock out, tag out procedures during service</li><li>Others – to specify (e.g., created SWP/RA document numbers)</li></ul> |
| 3D Printing                                                                                                                                                                                                                                              | Heated Nozzle and Fumes from ABS                                                                                                                                                                                                                   | Risk Controls as per RA (EDIC/RA/EW/026 & EDIC/RA/EW/036)<br>Standard RAs                                                                                                                                                                                               |
| Assembling Tool kit with hand tools                                                                                                                                                                                                                      | Improper usage of tools, damaged tools                                                                                                                                                                                                             | Risk Control as per RA (EDIC/RA/EW/086)<br>Standard RA                                                                                                                                                                                                                  |
| Assembling Robot with hand tools                                                                                                                                                                                                                         | Improper usage of tools, damaged tools                                                                                                                                                                                                             | Risk Control as per RA (EDIC/RA/EW/086)<br>Standard RA                                                                                                                                                                                                                  |
| Testing of Robot after assembly                                                                                                                                                                                                                          | Obstruction near robot<br>Malfunction                                                                                                                                                                                                              | To create Activity RA                                                                                                                                                                                                                                                   |

These are listed in EDIC workshops

## Chemical / Biological Hazards

| Type of Chemical/ Biological substances<br><small>(Please state clearly the type of chemical/ Biological substances to be used and how it is to be used during the project, including the handling, storage and disposal.)</small> | Hazard Description<br><small>(Please describe the kind of hazard which might be present)</small> | Risk Control Implemented<br><small>(or created RA document number)</small> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Usage of commercial off the shelf cleaning detergent                                                                                                                                                                               | Skin irritation                                                                                  | Frequent washing of hands after contact with detergent                     |
|                                                                                                                                                                                                                                    |                                                                                                  |                                                                            |

# Illustration (Project Risk Evaluation)

Activities / Experiments

| Will there be an experiment / test / project activity carried out? |                                                            | Yes / <del>No</del>                                                                              |
|--------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| S/No.                                                              | Activity / Experiment name                                 | Activity / Experiment Risk Assessment required<br>(state RA number or planned date to create RA) |
| 1                                                                  | Assembly and Testing of water pump                         | RA/CDE3301/IS399/2025-1 (Apr 2025)                                                               |
| 2                                                                  | Assembly and Testing of battery and electronics on Trolley | RA/CDE3301/IS399/2025-2 (Apr 2025)                                                               |
| 3                                                                  | Assembly and Testing of Pneumatic System                   | RA/CDE3301/IS399/2025-3 (May 2025)                                                               |
| 4                                                                  | Testing of Cleaning Robot                                  | RA/CDE3301/IS399/2025-4 (Jun 2025)                                                               |
|                                                                    |                                                            |                                                                                                  |

| Other Hazards                                                                                                                                                                                                                 |                                                                                                  |                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Type of Device/Tool/Machine<br><small>(If the device you use is not listed in the tools/machines registry from EDIC or the use of tool/machine deviates from the standard SWP provided by EDIC, please state it here)</small> | Hazard Description<br><small>(Please describe the kind of hazard which might be present)</small> | Risk Control Implemented                                |
| Usage of Soldering iron to solder components                                                                                                                                                                                  | Hot soldering iron and soldering fumes                                                           | Risk Controls as per RA (EDIC/RA/EW/045)<br>Standard RA |
|                                                                                                                                                                                                                               |                                                                                                  |                                                         |
|                                                                                                                                                                                                                               |                                                                                                  |                                                         |

| Risk Summary                      |                                       |
|-----------------------------------|---------------------------------------|
| Identify the main Hazard          | Describe the main control             |
| Mechanical and Electrical hazards | Controls as per Activity RAs attached |

Declaration (Please tick one)

|                          |                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Risks posed is mild/moderate and unlikely to increase during the course of project  |
| <input type="checkbox"/> | Risks are significant but will be effectively controlled                            |
| <input type="checkbox"/> | There is uncertainty about the level of risks                                       |
| <input type="checkbox"/> | Risks are significant and cannot be effectively controlled                          |
| <input type="checkbox"/> | * For Chemical Hazards, please ensure that the SDS (Safety Data Sheet) is attached. |
| <input type="checkbox"/> | SDS Attachment (please tick if they are attached with this form)                    |
|                          | SDS Validity Period: <u>NA</u> to <u>NA</u> (please indicate the dates)             |

# Illustration (Project Risk Evaluation)

|                                      |                                                                                   |                                                                                    |                                                                                     |  |  |
|--------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--|--|
| Conducted by:<br>(Name, designation) | Tan Siew Peng                                                                     | Mohammad Ali                                                                       | <u>Raveen</u> Sharma                                                                |  |  |
| Signature:                           |  |  |  |  |  |
| Date:                                | 13 Feb 2025                                                                       | 13 Feb 2025                                                                        | 13 Feb 2025                                                                         |  |  |

|                                  |                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------|
| Approved by: (Name, designation) | Thomas Job (Project Supervisor)                                                    |
| Signature:                       |  |
| Date:                            | 14 Feb 2025                                                                        |

# Activity/Experiment based RA

# Activity or Experiment-based Risk Assessment (RA)

If you are conducting an experiment or carrying out a project activity, please use the Activity / Experiment-based RA form to conduct RA.

1) Fill up the tasks, hazards identified, consequences and risk controls.

| Activity or Experiment-Based Risk Assessment Form |  |                                 |  |  |  |                        |  |  |  |  |  |  |  |
|---------------------------------------------------|--|---------------------------------|--|--|--|------------------------|--|--|--|--|--|--|--|
| Laboratory:                                       |  | Name of Experiment/Activity:    |  |  |  | FYP/Project/Activities |  |  |  |  |  |  |  |
| Location:                                         |  | Name of student/RF/RE/RS/staff: |  |  |  | Name of PI/Supervisor: |  |  |  |  |  |  |  |
| Last Review Date:                                 |  | Next Review Date:               |  |  |  | Doc no.                |  |  |  |  |  |  |  |

| 1. Hazard Identification |      |         |                       |                                | 2. Risk Evaluation & Control |   |   |                               |   |   |   |           |          |
|--------------------------|------|---------|-----------------------|--------------------------------|------------------------------|---|---|-------------------------------|---|---|---|-----------|----------|
| No.                      | Task | Hazards | Possible Consequences | Existing Risk Control (if any) | S                            | L | R | Additional / New Risk Control | S | L | R | Action By | Deadline |
|                          |      |         |                       |                                |                              |   |   |                               |   |   |   |           |          |
|                          |      |         |                       |                                |                              |   |   |                               |   |   |   |           |          |
|                          |      |         |                       |                                |                              |   |   |                               |   |   |   |           |          |

# RA – Risk Controls

Each RA is valid for 3 years unless there is a need to review

| Activity or Experiment-Based Risk Assessment Form |                                          |                                    |                                          |                                                                                                                            |                              |   |   |                               |             |   |   |           |          |
|---------------------------------------------------|------------------------------------------|------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------|---|---|-------------------------------|-------------|---|---|-----------|----------|
| Department:                                       | Engineering Design and Innovation Centre |                                    |                                          | Name of Experiment/Activity:                                                                                               | Testing of a Drone Propeller |   |   |                               |             |   |   |           |          |
| Location:                                         | E2A-02-02                                |                                    |                                          | Name of Person in-charge:                                                                                                  | Tan Siew Peng                |   |   | Name of Supervisor:           | Thomas Jobs |   |   |           |          |
| Review Date:                                      | 3 Feb 2020                               |                                    |                                          | Next Review Date:                                                                                                          | 2 Feb 2023                   |   |   | Document No: EDIC/RA/EW/099   |             |   |   |           |          |
| 1. Hazard Identification                          |                                          |                                    |                                          | 2. Risk Evaluation & Control                                                                                               |                              |   |   |                               |             |   |   |           |          |
| No.                                               | Task                                     | Hazards                            | Possible Consequences                    | Existing Risk Control (if any)                                                                                             | S                            | L | R | Additional / New Risk Control | S           | L | R | Action By | Deadline |
| 1                                                 | Preparation for test                     | Accidental start-up                | Cuts from Propeller                      | To follow a checklist and assigned roles<br>Operator to be trained according to SWP<br>To implement an electronic cut off. | 2                            | 1 | 2 |                               |             |   |   |           |          |
|                                                   |                                          | Small sparks on connecting battery | Burning of connectors and injury to hand | Practice for fast and correct method of connection                                                                         | 1                            | 2 | 2 |                               |             |   |   |           |          |

For each task, identify the hazards (e.g., slippery floor due to water test) and possible consequence (e.g., injury due to fall) and put in the controls to prevent the consequence (e.g., keep the floor dry at all time)

# Hierarchy of Controls



# Examples of Risk Controls

**Elimination:** remove a flammable product; remove a tripping hazard; remove a sharp object, etc.

**Substitution:** replace a solvent with water-based solution; replace an experiment with simulation; replace vapor heating with electric heating; replace pneumatic control with electric control; replace a poisonous chemical with a less toxic one, etc.

**Engineering Controls:** Bio Safety Cabinet; Fume Hood; Exhaust Ventilation; Safety Guards; Interlocks; Secondary Containment; Safety Alarms; ELCB, etc.

**Administrative Controls:** SOP; Labels and Signages; Verification Tests; Vaccinations; Training and Education; Safety Inspection; Equipment Maintenance, etc.

**PPE:** Safety Helmets; Face masks; Safety Goggles; Ear Mufflers; Safety Gloves; Safety Shoes; Safety Vests/Coats, Safety Harnesses, etc.

# Risk Levels

| Activity or Experiment-Based Risk Assessment Form |                                          |                                    |                                          |                                                                                                                                                   |                              |   |   |                               |             |   |   |           |          |  |
|---------------------------------------------------|------------------------------------------|------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---|---|-------------------------------|-------------|---|---|-----------|----------|--|
| Department:                                       | Engineering Design and Innovation Centre |                                    |                                          | Name of Experiment/Activity:                                                                                                                      | Testing of a Drone Propeller |   |   |                               |             |   |   |           |          |  |
| Location:                                         | E2A-02-02                                |                                    |                                          | Name of Person in-charge:                                                                                                                         | Tan Siew Peng                |   |   | Name of Supervisor:           | Thomas Jobs |   |   |           |          |  |
| Review Date:                                      | 3 Feb 2020                               |                                    |                                          | Next Review Date:                                                                                                                                 | 2 Feb 2023                   |   |   | Document No: EDIC/RA/EW/099   |             |   |   |           |          |  |
| 1. Hazard Identification                          |                                          |                                    |                                          | 2. Risk Evaluation & Control                                                                                                                      |                              |   |   |                               |             |   |   |           |          |  |
| No.                                               | Task                                     | Hazards                            | Possible Consequences                    | Existing Risk Control (if any)                                                                                                                    | S                            | L | R | Additional / New Risk Control | S           | L | R | Action By | Deadline |  |
| 1                                                 | Preparation for test                     | Accidental start-up                | Cuts from Propeller                      | To follow a checklist and assigned roles<br>Operator to be trained according to SWP<br>To implement an electronic cut off.                        | 2                            | 1 | 2 |                               |             |   |   |           |          |  |
|                                                   |                                          | Small sparks on connecting battery | Burning of connectors and injury to hand | Practice for fast and correct method of connection                                                                                                | 1                            | 2 | 2 |                               |             |   |   |           |          |  |
| 2                                                 | Propeller start up                       | Flying debris                      | Cuts and injury by debris                | Area to be checked and cleared of debris<br>Put on Safety Goggles to protect eyes                                                                 | 2                            | 1 | 2 |                               |             |   |   |           |          |  |
|                                                   |                                          | Propeller shatter or giving way    | Cuts and injury by shattered blades      | Area for operation to be cleared<br>Put a physical barrier between system and operator<br>Put on Safety Goggles to protect eyes<br>Remote testing | 2                            | 1 | 2 |                               |             |   |   |           |          |  |

Severity   Likelihood   Risk

# Understanding “Severity (S)” and “Likelihood (L)”

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Likelihood</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                |
| <b>Unlikely (1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Not likely to occur (has not occurred in the PI's Lab or similar Lab setup.)                                                                                                                   |
| <b>Possible (2)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Possible or known to occur (has occurred in the PI's Lab or Similar Lab setup.)                                                                                                                |
| <b>Very Likely (3)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Common or repeating occurrence (has occurred repetitively in the PI's Lab or similar Lab setup.)                                                                                               |
| <b><u>Severity</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                |
| <b>Low (1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (e.g. No injury, injury or ill-health requiring first aid treatment only - includes minor cuts and bruises, irritation, ill-health with temporary discomfort)                                  |
| <b>Medium (2)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (e.g. Injury requiring medical treatment or ill-health leading to disability – includes lacerations, burns, sprains, minor fractures, dermatitis, deafness, work-related upper limb disorders) |
| <b>High (3)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (e.g. Fatal, serious injury or life-threatening occupational disease – includes amputations, major fractures, multiple injuries, occupational cancer, acute poisoning and fatal diseases)      |
| <b><u>Severity</u> - Consider the magnitude/severity of the consequences of the Risk Factor occurring and then list this as 3 (High), 2 (Moderate) or 1 (Low).</b>                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                |
| <b>Severity normally will not change unless there is a physical change to the equipment or process.</b>                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                |
| <b><u>Likelihood</u> - Team should rely upon their experience and consider realistic scenarios. Listed below are examples of factors that may be considered in determining the likelihood.</b>                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                |
| <ul style="list-style-type: none"><li>- Past experience / incidents</li><li>- Complexity of the activity</li><li>- Number of personnel involved in the activity (e.g. all personnel, a limited number of trained personnel, etc)</li><li>- Frequency of use or execution</li><li>- Degree of control (involvement of contractors)</li><li>- Strength/completeness of administrative controls</li><li>- Sufficiency/formality of training</li><li>- Others....</li></ul> |                                                                                                                                                                                                |

# Activity or Experiment-based Risk Assessment (RA)

- 2) Refer to the RA form for guidelines on deriving the Severity (S) and Likelihood (L) levels.
- 3) Assess the Severity (S) and Likelihood (L) of each consequence and derive the respective Risk level (R) according to the Risk Matrix shown below:

|          |      | Likelihood |          |          |
|----------|------|------------|----------|----------|
|          |      | Likely     | Possibly | Unlikely |
| Severity | Low  | 3          | 2        | 1        |
|          | Med  | 6          | 4        | 2        |
|          | High | 9          | 6        | 3        |

| Risk = Likelihood x Severity |                                  |
|------------------------------|----------------------------------|
| RISK                         | DECISION PROCESS                 |
| < 3                          | RISK ACCEPTABLE                  |
| 3, 4                         | CONSIDER ADDITIONAL RISK CONTROL |
| > 4                          | ADDITIONAL RISK CONTROL REQUIRED |

- 4) If the Risk level is assessed to be above 4, do not proceed with the experiment / activity. Relook into the controls so that the Risk level can be brought down to 4 and below.
- 5) If you have a Risk Level of 3 or 4, the Risk Control must include Engineering Controls and PPEs.
- 6) A Risk level of 2 and below is recommended.

# Illustration (Activity RA)

Conduct an Activity RA for every Activity identified in the Project RA

Activities / Experiments

| Will there be an experiment / test / project activity carried out? |                                                            | Yes / <del>No</del>                                                                           |
|--------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| S/No.                                                              | Activity / Experiment name                                 | Activity / Experiment Risk Assessment required (state RA number or planned date to create RA) |
| 1                                                                  | Assembly and Testing of water pump                         | RA/CDE3301/IS399/2025-1 (Apr 2025)                                                            |
| 2                                                                  | Assembly and Testing of battery and electronics on Trolley | RA/CDE3301/IS399/2025-2 (Apr 2025)                                                            |
| 3                                                                  | Assembly and Testing of Pneumatic System                   | RA/CDE3301/IS399/2025-3 (May 2025)                                                            |
| 4                                                                  | Testing of Cleaning Robot                                  | RA/CDE3301/IS399/2025-4 (Jun 2025)                                                            |

Other Hazards

| Type of Device/Tool/Machine<br>(If the device you use is not listed in the tools/machines registry from EDIC or the use of tool/machine deviates from the standard SWP provided by EDIC, please state it here) | Hazard Description<br>(Please describe the kind of hazard which might be present) | Risk Control Implemented                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------|
| Usage of Soldering iron to solder components                                                                                                                                                                   | Hot soldering iron and soldering fumes                                            | Risk Controls as per RA (EDIC/RA/EW/045) Standard RA |
|                                                                                                                                                                                                                |                                                                                   |                                                      |
|                                                                                                                                                                                                                |                                                                                   |                                                      |

Risk Summary

| Identify the main Hazard          | Describe the main control             |
|-----------------------------------|---------------------------------------|
| Mechanical and Electrical hazards | Controls as per Activity RAs attached |

Declaration (Please tick one)

|                          |                                                                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Risks posed is mild/moderate and unlikely to increase during the course of project                                                                                                                                                         |
| <input type="checkbox"/> | Risks are significant but will be effectively controlled                                                                                                                                                                                   |
| <input type="checkbox"/> | There is uncertainty about the level of risks                                                                                                                                                                                              |
| <input type="checkbox"/> | Risks are significant and cannot be effectively controlled                                                                                                                                                                                 |
| <input type="checkbox"/> | For Chemical Hazards, please ensure that the SDS (Safety Data Sheet) is attached.<br>SDS Attachment (please tick if they are attached with this form)<br>SDS Validity Period: _____ NA _____ to _____ NA _____ (please indicate the dates) |

# Illustration (Activity RA)

|   |                                |                                        |                                                                                                                                                                                      |
|---|--------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | <u>JDP</u> Assembly Workshop 1 | Assembly & Testing of pneumatic system | Connect tubes for compressed air tank, solenoid valve and pneumatic tool changer.<br>Connect wiring for solenoid valve to microcontroller<br>Charge up air tank<br>Testing of system |
|---|--------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



Location



Activity



Tasks and Sequence

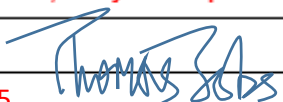
# Illustration (Activity RA)

| Activity or Experiment-Based Risk Assessment Form |                         |  |  |  |                                 |  |  |  |                                          |  |                                 |  |             |  |                        |
|---------------------------------------------------|-------------------------|--|--|--|---------------------------------|--|--|--|------------------------------------------|--|---------------------------------|--|-------------|--|------------------------|
| Laboratory:                                       | IDP Assembly Workshop 1 |  |  |  | Name of Experiment/Activity:    |  |  |  | Assembly and Testing of Pneumatic System |  |                                 |  |             |  | FYP/Project/Activities |
| Location:                                         | E1A-01-13               |  |  |  | Name of student/RF/RE/RS/staff: |  |  |  | Mohammad Ali                             |  | Name of PI/Supervisor:          |  | Thomas Jobs |  |                        |
| Last Review Date:                                 | 02/05/2025              |  |  |  | Next Review Date:               |  |  |  | 01/05/2028                               |  | Doc no. RA/CDE3301/IS399/2025-3 |  |             |  |                        |

| 1. Hazard Identification |                                                                                   |                 |                                              |                                                                                                                                                                                                     | 2. Risk Evaluation & Control |   |   |                               |   |   |   |           |          |  |
|--------------------------|-----------------------------------------------------------------------------------|-----------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---|---|-------------------------------|---|---|---|-----------|----------|--|
| No.                      | Task                                                                              | Hazards         | Possible Consequences                        | Existing Risk Control (if any)                                                                                                                                                                      | S                            | L | R | Additional / New Risk Control | S | L | R | Action By | Deadline |  |
| 1                        | Connect tubes for compressed air tank, solenoid valve and pneumatic tool changer. | Pressurized air | Eye and face injuries by pressurised air     | Ensure all air connectors are tightened with wrench<br><br>Wear PPE such as goggles or face shield                                                                                                  | 2                            | 1 | 2 |                               |   |   |   |           |          |  |
| 2                        | Connect wiring for solenoid valve to microcontroller                              | Short circuit   | Electric shock to humans                     | Check condition of wires and connectors before turning on power<br><br>Keep equipment and human dry and clean<br><br>Ensure proper grounding<br><br>Check and ensure voltage rating to power supply | 2                            | 1 | 2 |                               |   |   |   |           |          |  |
| 3                        | Charge up air tank                                                                | Pressurized air | Injuries from explosion of overcharged tank. | Ensure all air sockets seals are tightened with wrench.                                                                                                                                             | 3                            | 1 | 3 |                               |   |   |   |           |          |  |

# Illustration (Activity RA)

|   |                   |               |                          |                                                                                                                                                                                                     |   |   |   |  |  |  |  |  |  |
|---|-------------------|---------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|--|--|--|--|--|--|
|   |                   |               |                          | Charge pressure below maximum allowed pressure of the tank.<br><br>Wear face shield/goggles                                                                                                         |   |   |   |  |  |  |  |  |  |
| 4 | Testing of system | Short circuit | Electric shock to humans | Check condition of wires and connectors before turning on power<br><br>Keep equipment and human dry and clean<br><br>Ensure proper grounding<br><br>Check and ensure voltage rating to power supply | 2 | 1 | 2 |  |  |  |  |  |  |

|                                                                |                                                                                      |                                                                                       |                                  |                                                                                       |  |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------|--|
| Conducted by: (Name, designation)                              | Mohammad Ali, Student                                                                |                                                                                       | Approved by: (Name, designation) | Thomas Jobs, Project Supervisor                                                       |  |
| Signature:                                                     |  |                                                                                       | Signature:                       |  |  |
| Date:                                                          | 01/05/2025                                                                           |                                                                                       | Date:                            | 02/05/2025                                                                            |  |
| Conducted in the presence of: (name of student/RF/RE/RS/staff) | Raveen Sharma                                                                        | Tan Siew Peng                                                                         |                                  |                                                                                       |  |
| Signature:                                                     |   |  |                                  |                                                                                       |  |
| Date:                                                          | 01/05/2025                                                                           | 01/05/2025                                                                            |                                  |                                                                                       |  |

# **Further Notes on Severity, Likelihood and Risks**

Severity and Likelihood are not constant. They may change with time and situation.

So do Risk levels.

# Illustration (Activity RA)

Illustration of Activity RA during early 2020, when Covid-19 was just discovered.

| 1. Hazard Identification |                                        |          |                          | 2. Risk Evaluation & Control   |   |   |   |                               |   |   |   |           |          |
|--------------------------|----------------------------------------|----------|--------------------------|--------------------------------|---|---|---|-------------------------------|---|---|---|-----------|----------|
| No.                      | Task                                   | Hazards  | Possible Consequences    | Existing Risk Control (if any) | S | L | R | Additional / New Risk Control | S | L | R | Action By | Deadline |
| 4                        | Testing module A in ABC Hospital wards | COVID-19 | Infected by the COVID-19 | Mask, temperature taking       | 3 | 3 | 9 | Do not proceed with testing   | - | - | - |           |          |

|          |      | Likelihood |          |          |
|----------|------|------------|----------|----------|
|          |      | Likely     | Possibly | Unlikely |
| Severity | Low  | 3          | 2        | 1        |
|          | Med  | 6          | 4        | 2        |
|          | High | 9          | 6        | 3        |

Risk = Likelihood x Severity

| RISK | DECISION PROCESS                 |
|------|----------------------------------|
| < 3  | RISK ACCEPTABLE                  |
| 3, 4 | CONSIDER ADDITIONAL RISK CONTROL |
| > 4  | ADDITIONAL RISK CONTROL REQUIRED |

| Likelihood                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unlikely                                                                                                                                                                                                                                                                                                                                                                                                                                                | Not likely to occur (has not occurred in the PI's Lab or similar Lab setup.)                                                                                                                   |
| Possible                                                                                                                                                                                                                                                                                                                                                                                                                                                | Possible or known to occur (has occurred in the PI's Lab or Similar Lab setup.)                                                                                                                |
| Very Likely                                                                                                                                                                                                                                                                                                                                                                                                                                             | Common or repeating occurrence (has occurred repetitively in the PI's Lab or similar Lab setup.)                                                                                               |
| Severity                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                |
| Low                                                                                                                                                                                                                                                                                                                                                                                                                                                     | (e.g. No injury, injury or ill-health requiring first aid treatment only - includes minor cuts and bruises, irritation, ill-health with temporary discomfort)                                  |
| Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                  | (e.g. Injury requiring medical treatment or ill-health leading to disability – includes lacerations, burns, sprains, minor fractures, dermatitis, deafness, work-related upper limb disorders) |
| High                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (e.g. Fatal, serious injury or life-threatening occupational disease – includes amputations, major fractures, multiple injuries, occupational cancer, acute poisoning and fatal diseases)      |
| <b>Severity</b> - Consider the magnitude/severity of the consequences of the Risk Factor occurring and then list this as 3 (High), 2 (Moderate) or 1 (Low).<br>Severity normally will not change unless there is a physical change to the equipment or process.                                                                                                                                                                                         |                                                                                                                                                                                                |
| <b>Likelihood</b> - Team should rely upon their experience and consider realistic scenarios. Listed below are examples of factors that may be considered in determining the likelihood.                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                |
| <ul style="list-style-type: none"><li>Past experience / incidents</li><li>Complexity of the activity</li><li>Number of personnel involved in the activity (e.g. all personnel, a limited number of trained personnel, etc)</li><li>Frequency of use or execution</li><li>Degree of control (involvement of contractors)</li><li>Strength/completeness of administrative controls</li><li>Sufficiency/formality of training</li><li>Others....</li></ul> |                                                                                                                                                                                                |

# Illustration (Activity RA)

What about early 2022? Situation has changed. Omicron symptoms are mild although common.

| 1. Hazard Identification |                                        |          |                          | 2. Risk Evaluation & Control                                               |   |   |   |                               |   |   |   |           |          |
|--------------------------|----------------------------------------|----------|--------------------------|----------------------------------------------------------------------------|---|---|---|-------------------------------|---|---|---|-----------|----------|
| No.                      | Task                                   | Hazards  | Possible Consequences    | Existing Risk Control (if any)                                             | S | L | R | Additional / New Risk Control | S | L | R | Action By | Deadline |
| 4                        | Testing module A in ABC Hospital wards | COVID-19 | Infected by the COVID-19 | Mask Fitting, Minimise contact time to 1 hour<br>Follow Hospital Protocols | 1 | 3 | 3 |                               | - | - | - |           |          |

|          |      | Likelihood |          |          |
|----------|------|------------|----------|----------|
|          |      | Likely     | Possibly | Unlikely |
| Severity | Low  | 3          | 2        | 1        |
|          | Med  | 6          | 4        | 2        |
|          | High | 9          | 6        | 3        |

| Risk = Likelihood x Severity |                                  |
|------------------------------|----------------------------------|
| RISK                         | DECISION PROCESS                 |
| < 3                          | RISK ACCEPTABLE                  |
| 3, 4                         | CONSIDER ADDITIONAL RISK CONTROL |
| > 4                          | ADDITIONAL RISK CONTROL REQUIRED |

| Likelihood                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unlikely                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Not likely to occur (has not occurred in the PI's Lab or similar Lab setup.)                                                                                                                   |
| Possible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Possible or known to occur (has occurred in the PI's Lab or Similar Lab setup.)                                                                                                                |
| Very Likely                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Common or repeating occurrence (has occurred repetitively in the PI's Lab or similar Lab setup.)                                                                                               |
| Severity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                |
| Low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | (e.g. No injury, injury or ill health requiring first aid treatment only - includes minor cuts and bruises, irritation, ill-health with temporary discomfort)                                  |
| Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (e.g. injury requiring medical treatment or ill-health leading to disability – includes lacerations, burns, sprains, minor fractures, dermatitis, deafness, work-related upper limb disorders) |
| High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (e.g. Fatal, serious injury or life-threatening occupational disease – includes amputations, major fractures, multiple injuries, occupational cancer, acute poisoning and fatal diseases)      |
| <b>Severity</b> - Consider the magnitude/severity of the consequences of the Risk Factor occurring and then list this as 3 (High), 2 (Moderate) or 1 (Low).<br>Severity normally will not change unless there is a physical change to the equipment or process.                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                |
| <b>Likelihood</b> - Team should rely upon their experience and consider realistic scenarios. Listed below are examples of factors that may be considered in determining the likelihood. <ul style="list-style-type: none"><li>- Past experience / incidents</li><li>- Complexity of the activity</li><li>- Number of personnel involved in the activity (e.g. all personnel, a limited number of trained personnel, etc)</li><li>- Frequency of use or execution</li><li>- Degree of control (involvement of contractors)</li><li>- Strength/completeness of administrative controls</li><li>- Sufficiency/formality of training</li><li>- Others....</li></ul> |                                                                                                                                                                                                |

Every project member is to sign on the last page and the project supervisor is to approve the RA before project may commence.

Project members to sign

Project supervisor to sign

|                                   |                                 |                                |                                 |                                  |                                  |  |  |  |  |  |  |  |  |  |  |
|-----------------------------------|---------------------------------|--------------------------------|---------------------------------|----------------------------------|----------------------------------|--|--|--|--|--|--|--|--|--|--|
|                                   |                                 |                                |                                 |                                  |                                  |  |  |  |  |  |  |  |  |  |  |
| Conducted by: (Name, designation) | Tan Siew Peng<br>Project Member | Mohammed Ali<br>Project Member | Raveen Sharma<br>Project Member | Approved by: (Name, designation) | Thomas Job<br>Project Supervisor |  |  |  |  |  |  |  |  |  |  |
| Signature:                        |                                 |                                |                                 | Signature:                       |                                  |  |  |  |  |  |  |  |  |  |  |
| Date:                             |                                 |                                |                                 | Date:                            |                                  |  |  |  |  |  |  |  |  |  |  |

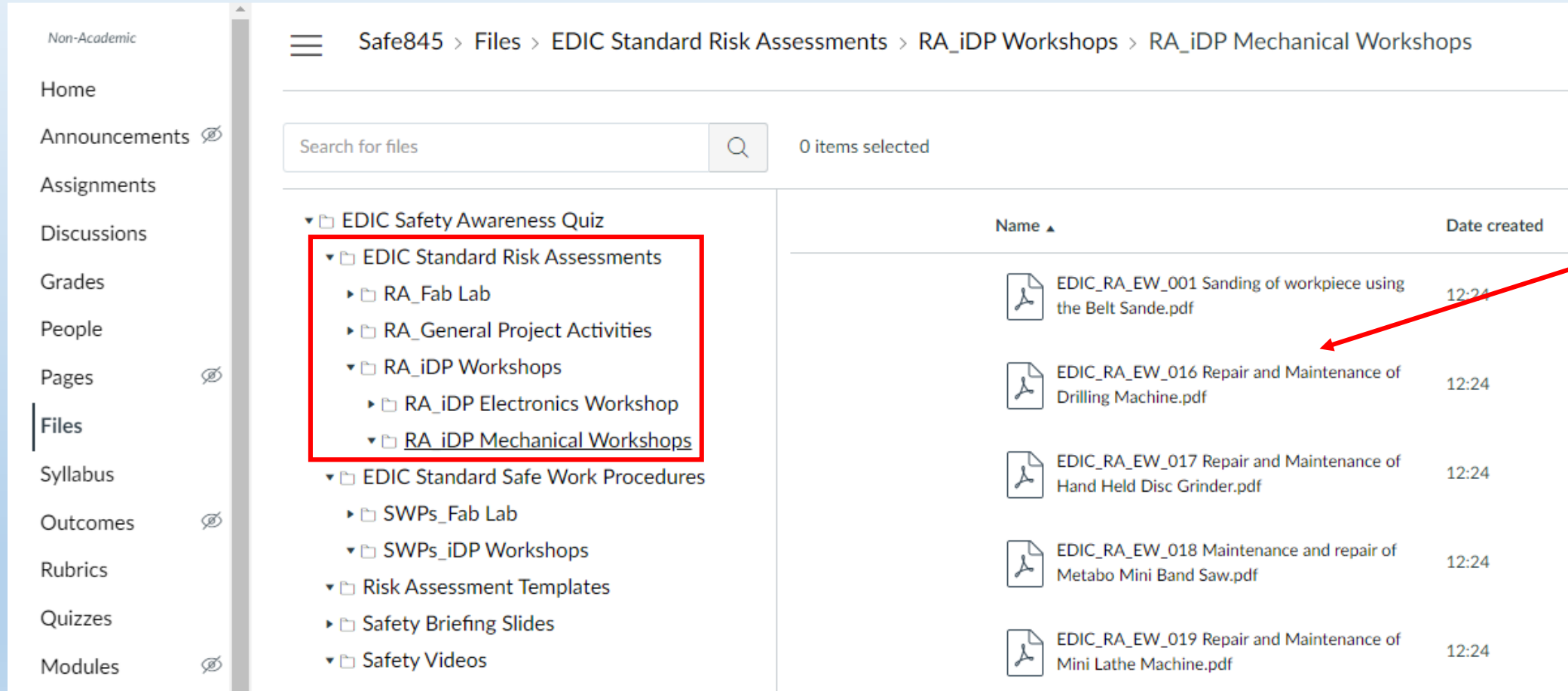
Fill in the date

Fill in the date of approval

## Where to access the EDIC standard Equipment Risk Assessments (RAs)?

**Via NUS Canvas:** Safe845 - Files – EDIC Safety Awareness Quiz - EDIC Standard Risk Assessments

**Via NUS Canvas:** iDP e-Station – Safety – Files – Standard Risk Assessments



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Safe845 > Files > EDIC Standard Risk Assessments > RA\_iDP Workshops > RA\_iDP Mechanical Workshops

Search for files

0 items selected

- EDIC Safety Awareness Quiz
  - EDIC Standard Risk Assessments
    - RA\_Fab Lab
    - RA\_General Project Activities
    - RA\_iDP Workshops
      - RA\_iDP Electronics Workshop
      - RA\_iDP Mechanical Workshops
    - EDIC Standard Safe Work Procedures
      - SWPs\_Fab Lab
      - SWPs\_iDP Workshops
    - Risk Assessment Templates
    - Safety Briefing Slides
    - Safety Videos

| Name                                                                | Date created |
|---------------------------------------------------------------------|--------------|
| EDIC_RA_EW_001 Sanding of workpiece using the Belt Sande.pdf        | 12:24        |
| EDIC_RA_EW_016 Repair and Maintenance of Drilling Machine.pdf       | 12:24        |
| EDIC_RA_EW_017 Repair and Maintenance of Hand Held Disc Grinder.pdf | 12:24        |
| EDIC_RA_EW_018 Maintenance and repair of Metabo Mini Band Saw.pdf   | 12:24        |
| EDIC_RA_EW_019 Repair and Maintenance of Mini Lathe Machine.pdf     | 12:24        |

Available RAs  
for your  
reference and  
adoption

## **When to submit the Risk Assessment?**

- 1) Submit before you start your hands on work of your project
- 2) Make sure the RA has a unique document number
- 3) Make sure the RA is signed by all project members and endorsed by the project PS
- 4) Save a scanned softcopy of your RA
- 5) You may need to submit the scanned RA softcopy with the document number to Lab officers for workshop/lab registration or through Canvas (your module) as required by your module coordinator

# Safe Working Procedures (SWPs)

|                                                                                                                                |                                           |                                                 |                      |                        |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------|----------------------|------------------------|
|  <b>NUS</b><br>National University of Singapore |                                           | <b>Engineering Design and Innovation Centre</b> | Document No.:        | EDIC/SWP/EW/009        |
| <b>Safe Work Procedure</b>                                                                                                     | <b>Operation of Laser Cutting Machine</b> |                                                 |                      | <b>No. of Pages: 3</b> |
| <b>Prepared by:</b>                                                                                                            | <b>Approved by:</b>                       | <b>Revision No.</b>                             | <b>Revision date</b> |                        |
| Ms. Annie Tan/<br>Miss Iris Ng                                                                                                 | Dr Liaw Hwee Choo                         | 1.0                                             | 20/11/2013           |                        |

### 1. OBJECTIVE

This procedure outlines the operation of the Epilog Laser Cutting Machine located in EDIC.

### 2. SCOPE

This procedure is applicable to DCP undergraduate students requiring the use and operation of the Epilog Laser Cutting Machine.

### 3. RESPONSIBILITY

- It is the responsibility of the Project Supervisor(s) in conjunction with the Laboratory Supervisor/ Laboratory in-charge / Laboratory in-charge of safety to ensure that all users using this procedure are adequately trained.
- It is the responsibility of the Laboratory technician to ensure that the Epilog Laser Cutting Machine is in a good working condition.
- It is the responsibility of the user to acquire the sufficient knowledge and training on the Epilog Laser Printing Machine before operation.

### 4. PROCEDURES

Before using the Epilog Laser Cutting,

- Students are required to ensure that the required parts are accurately drawn on CAD and the file should be saved in dxf format.
- The following should be checked
  - the overall size of the part to be printed and ensure that it does not exceed the specifications of the printer(24" by 18");
  - the scale of the CAD drawing
  - the drawing must be in the projection which the part is to be cut

Using the Epilog Laser Cutting,

A. Operating the program:

- On the mains of the all the machines and the computer
- Switch on the computer first, followed by the Fume Extractor System, Generator and lastly the Epilog Laser Cutter.
- Once the computer is in operation, launch the program "Corel Draw"

|                                                                                                                                  |                                           |                                                 |                      |                        |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------|----------------------|------------------------|
|  <b>NUS</b><br>National University of Singapore |                                           | <b>Engineering Design and Innovation Centre</b> | Document No.:        | EDIC/SWP/EW/009        |
| <b>Safe Work Procedure</b>                                                                                                       | <b>Operation of Laser Cutting Machine</b> |                                                 |                      | <b>No. of Pages: 3</b> |
| <b>Prepared by:</b>                                                                                                              | <b>Approved by:</b>                       | <b>Revision No.</b>                             | <b>Revision date</b> |                        |
| Ms. Annie Tan/<br>Miss Iris Ng                                                                                                   | Dr Liaw Hwee Choo                         | 1.0                                             | 20/11/2013           |                        |

- Select a new project sheet document "Epilog 24" x 18" as this is the default effective size which the laser is able to cut.
- Make sure that the units are in mm.
- Once the project sheet has been created, import the DXF file from your external storage device (such as thumb drives)
- Import the DXF as a 1:1 scale onto the sheet in Corel Draw.
- Check that the drawing is the item which you would like to laser.
- Once the item is verified, for cut lines, ensure that the lines are "hairline" in the line properties in the program and if there are areas to be engraved, the thickness of the lines can be adjusted in the line properties.
- Place the drawing to be worked on the top left hand corner for alignment.
- Click on "Ctrl + P" for the print job to be executed.
- Select the Epilog Laser Cutter and click on "Preferences" to set the speed and power according to the material which you wish to work on as indicated on the specs sheet.
- On the left bottom, select the Auto Focus option.
- Once the appropriate settings are done, send the print job

### B. Aligning the laser before cut

- The print job will be sent to the Epilog Laser Cutter and it will be reflected as a Job.
- Press the "Focus" button and press the down button to move the table of the laser cutter down. This will allow for the workpiece to be place on the table. The distance to be moved down is to be approximately 4x the thickness of the material to be used.
- Once the height is adjusted, press the "RESET" button to allow the laser to return to its original X-Y position. Note that the laser should be hit the workpiece when the laser is moving back to its original position. If it does, then the height of the table is then lowered automatically to the appropriate height.

### C. Executing the job

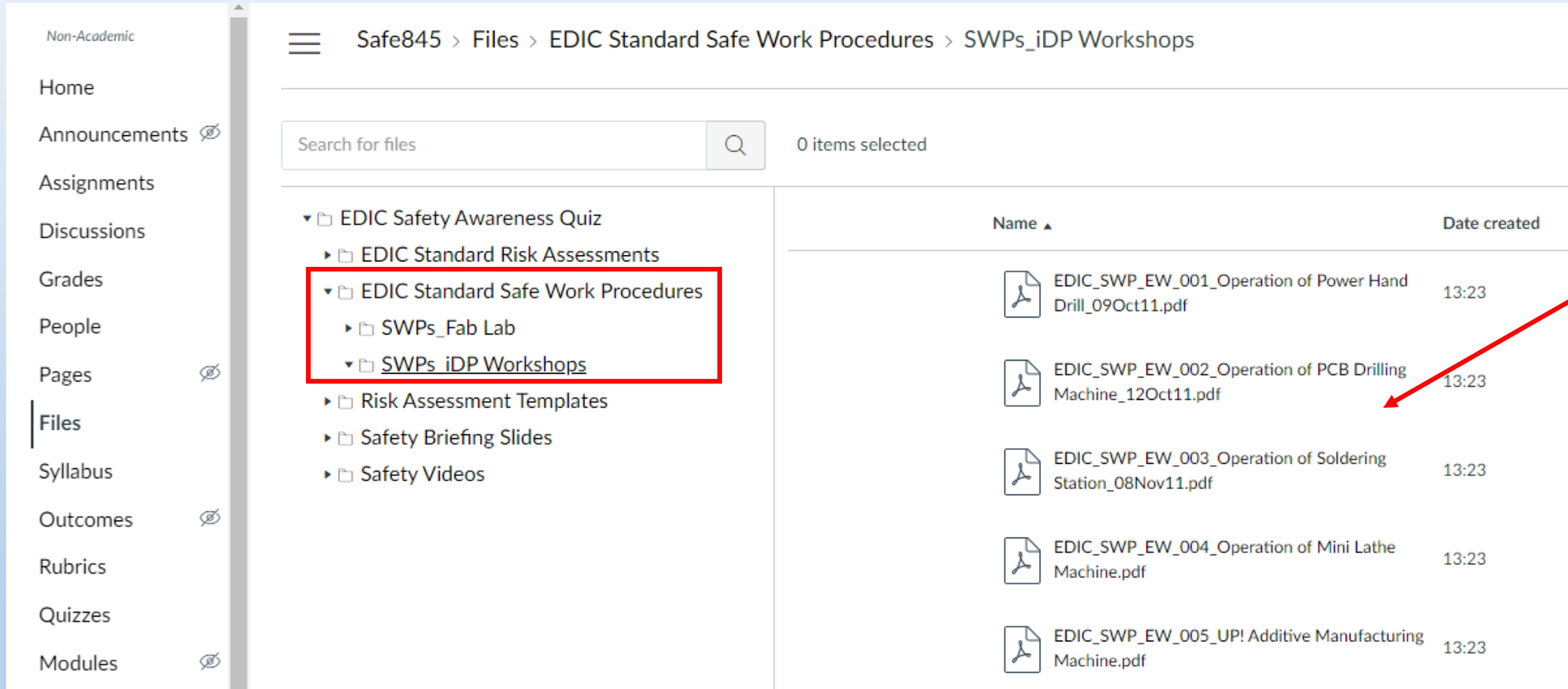
- Once the alignment is done, the job is ready to be executed
- On the "Pointer" button. If you wish to check if your workpiece is placed in the right position, execute the job without the top door close and the pointer button on. The red LED light will move and you can check if the workpiece is in the right position for the cut/engrave.
- Once all the checks are done, close the top door and execute the job.

## Example of Safe Work Procedure

# Where to access the EDIC standard Equipment Safe Work Procedure (SWPs)?

**Via NUS Canvas:** Safe845 - Files – EDIC Safety Awareness Quiz - EDIC Standard Safe Work Procedures

**Via NUS Canvas:** iDP e-Station – Safety – Files – Standard Safe Work Procedures



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  - Safety Briefing Slides
  - Safety Videos

| Name                                                          | Date created |
|---------------------------------------------------------------|--------------|
| EDIC_SWP_EW_001_Operation of Power Hand Drill_09Oct11.pdf     | 13:23        |
| EDIC_SWP_EW_002_Operation of PCB Drilling Machine_12Oct11.pdf | 13:23        |
| EDIC_SWP_EW_003_Operation of Soldering Station_08Nov11.pdf    | 13:23        |
| EDIC_SWP_EW_004_Operation of Mini Lathe Machine.pdf           | 13:23        |
| EDIC_SWP_EW_005_UP! Additive Manufacturing Machine.pdf        | 13:23        |

Available SWPs  
for your  
reference

# Mandatory Eye Protection

- 1) The University has extended the mandatory eye protection policy to teaching labs and workshops with effect from 1<sup>st</sup> Jan 2017
- 2) In particular, Safety goggle is required to be worn when working with materials of a medium hazard rating, for example, when handling chemical and biological agents or injurious materials such as projectiles (for example, when testing propellers) with a moderate possibility of splash or fly-off.
- 3) More information can be found at [https://inetapps.nus.edu.sg/osh/portal/genlab\\_safety/eyeprotection.html](https://inetapps.nus.edu.sg/osh/portal/genlab_safety/eyeprotection.html)

# Mandatory Eye Protection Policy

|   | Safety Glasses                                                                                                                                                                                                                                                                                                                                    | Safety Goggles<br>(Medium Risk Activities)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Face Shield + Safety glasses/goggles<br>(High Risk Activities)                                                                                                                                                                                                                                                                                                                                                                                                    |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p>The minimum requirement when in teaching and research laboratories and workshops where hazardous materials and/or activities are present.</p>                                                                                                                                                                                                  | <p>Required when:</p> <p>Working with materials that have a medium hazard rating<sup>2</sup> (corrosive, injurious<sup>1</sup>, infectious etc.) and a <u>moderate</u> splash probability exists.</p> <p>Examples of activities where safety goggles are required :</p> <ul style="list-style-type: none"><li>Mixing or pouring of corrosive/injurious materials out from stock bottle on the benchtop</li></ul>                                                                                                                           | <p>Required when:</p> <p>Working with materials that have a medium or high hazard rating (corrosive, injurious, infectious etc.) and a <u>high</u> splash probability exists.</p> <p>Examples of activities where face shield is needed together with safety glasses/goggles:</p> <ul style="list-style-type: none"><li>Conducting a reaction involving corrosive/injurious/infectious materials under pressure</li><li>Mixing or pouring of cryogenics</li></ul> |
| 2 | <p><b>Information on Safety Glasses</b></p>  <p>Safety glasses are similar to normal glasses but have lens that are impact resistant and frames that are much stronger. However, they do not provide adequate protection from significant chemical splashes.</p> | <p><b>Information on Safety Goggles</b></p>  <p>Like safety glasses, goggles are impact resistant. Safety goggles offer greater protection to the eye than safety glasses against chemical splashes. Safety goggles with indirect ventilation will prevent substances from draining into the eye. Some may be worn over prescription glasses. They are suitable when working with corrosive or injurious materials and a splash probability exists.</p> | <p><b>Information on Face Shield</b></p>  <p>Face shields provide splash protection to the wearer's entire face. They are suitable when working with corrosive or injurious materials and a high splash probability exists. Face shield shall be worn with safety goggles or safety glasses.</p>                                                                               |

<sup>1</sup> Injurious materials include physical hazards such as projectiles.

<sup>2</sup> Refer to the relevant SDS for information on the hazard(s) posed by the material being handled.

## **NUS Safety Policy on the usage of the following electrical parts:**

- 1) Electrical Plugs
- 2) Multi-plug adaptors
- 3) Universal Travel Adaptors
- 4) Extension cord

# Electrical Plugs and Sockets

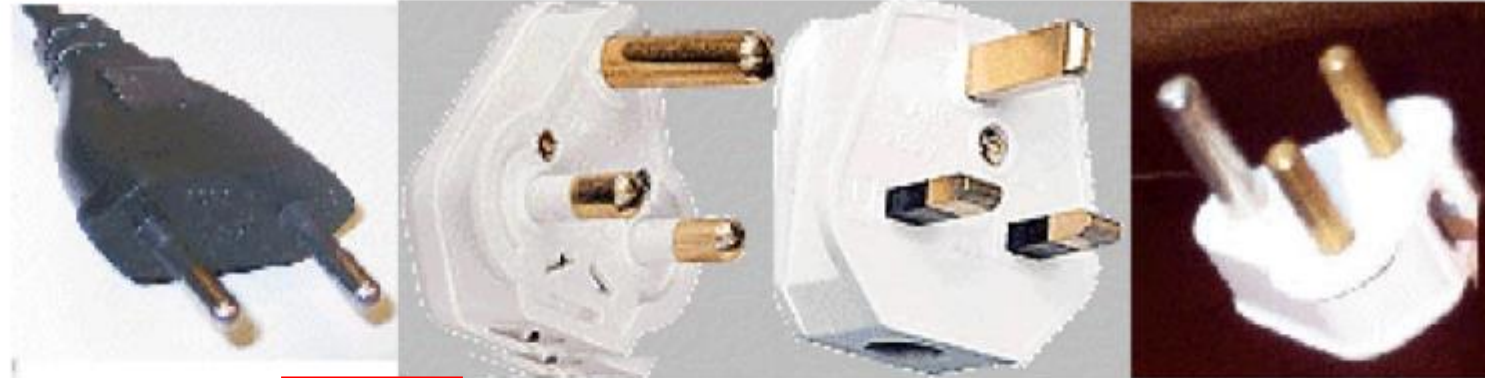


Figure 1: Type of plugs **suitable** for the socket outlets in NUS (From left: 2.5A round pin, 5A round pin, 13A rectangular pin, 15A round pin). Courtesy of Enterprise Singapore



Figure 2: Type of plugs **not suitable** for the socket outlets in NUS (From left : Flat-blade plug, Schuko plug with side grounding contact, Schuko plug with side grounding contact, 10/16-Amp round pin, Flat-blade but with round grounding pin, Flat-blade but with round grounding pin). Courtesy of Enterprise Singapore.

# Electrical Plugs and Sockets

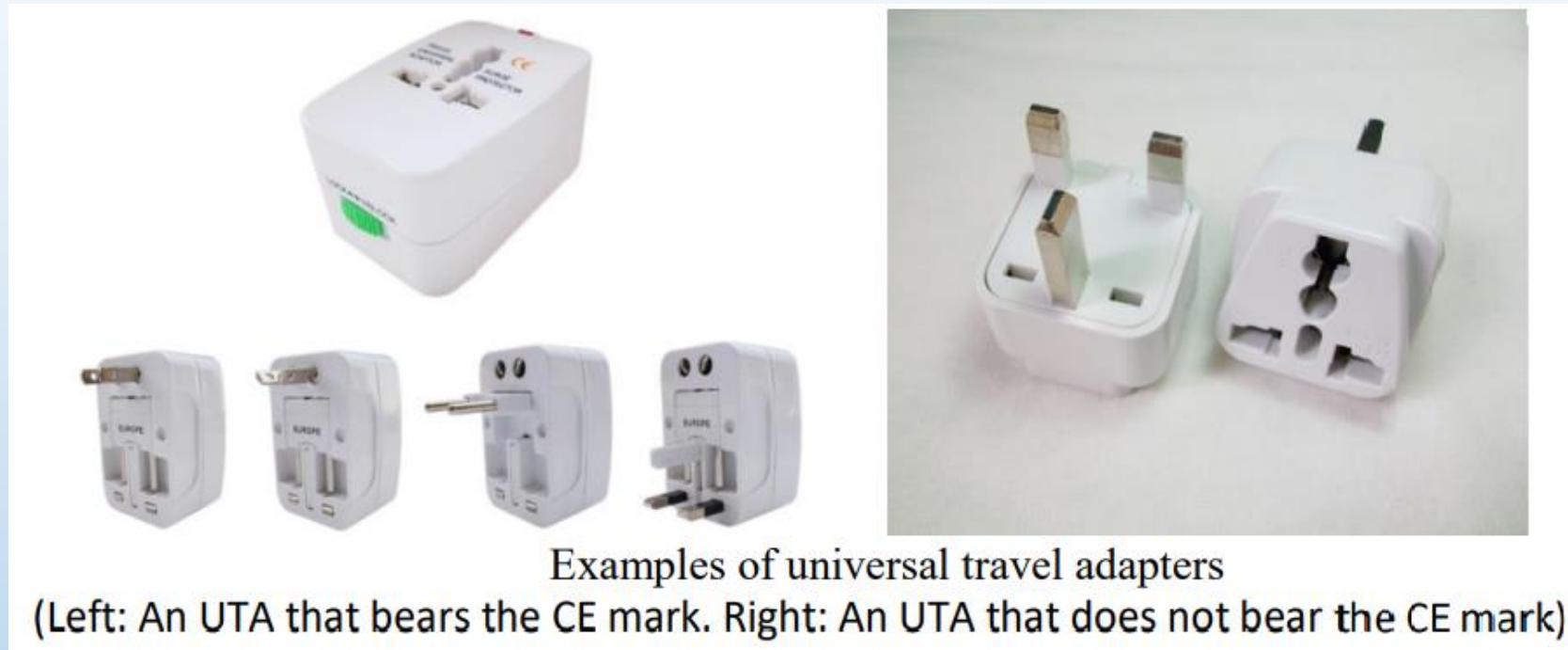
If you are unable to purchase an equipment with the acceptable plugs, you may engage own Licensed Electrical Worker (LEW) to change it to acceptable ones. You may also approach Office of Facilities Management (OFM) to help with the change.

# Usage of Electrical Adaptors



- Multi-plug adapters shall be labelled with the “Safety Mark” logo.
- All plugs inserted in a multi-plug adapter should be labelled with the respective current rating (in Amperes) of the equipment.
- The current rating of the equipment is usually listed on the equipment. Alternatively, it can be obtained by dividing the power rating of the equipment in Watts by 220 Volts.

# Usage of Universal Travel Adaptors

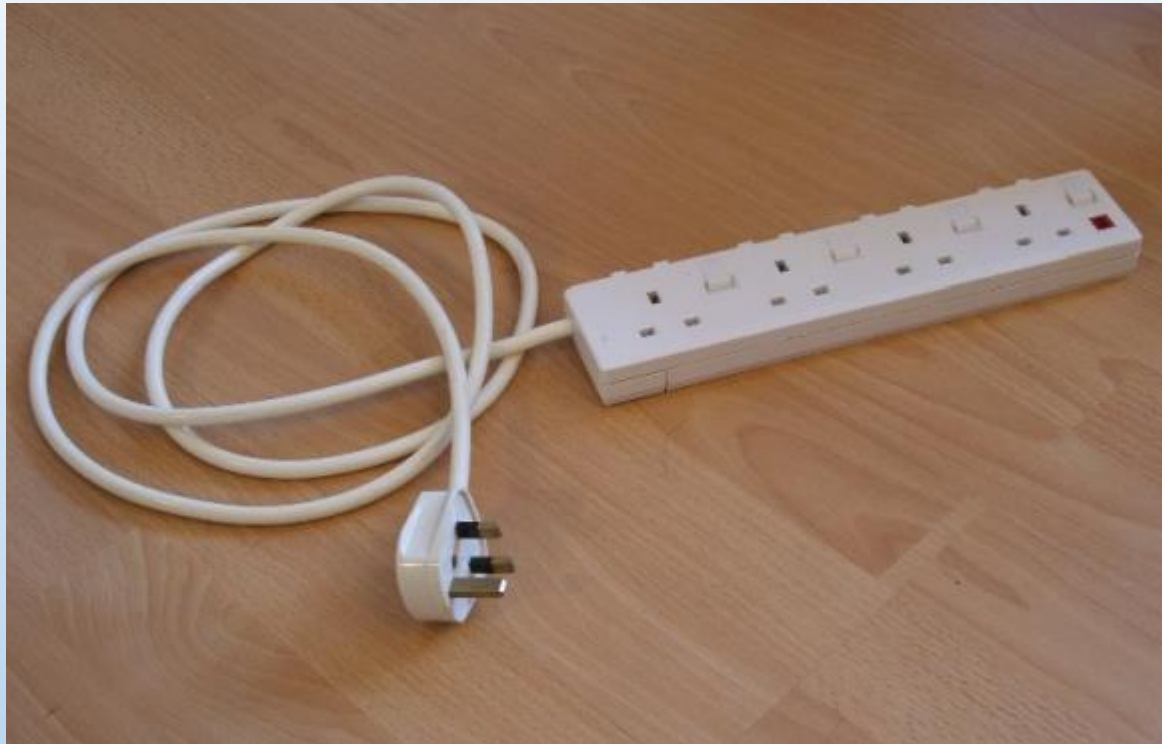


Only equipment that are rated to operate at 220 V shall be plugged into the Universal Travel Adaptors. Equipment that is plugged into the electrical socket for electrical power supply via a universal travel adapter (Figure 5) shall not exceed the adapter's amperage limit that is specified by the manufacturer.

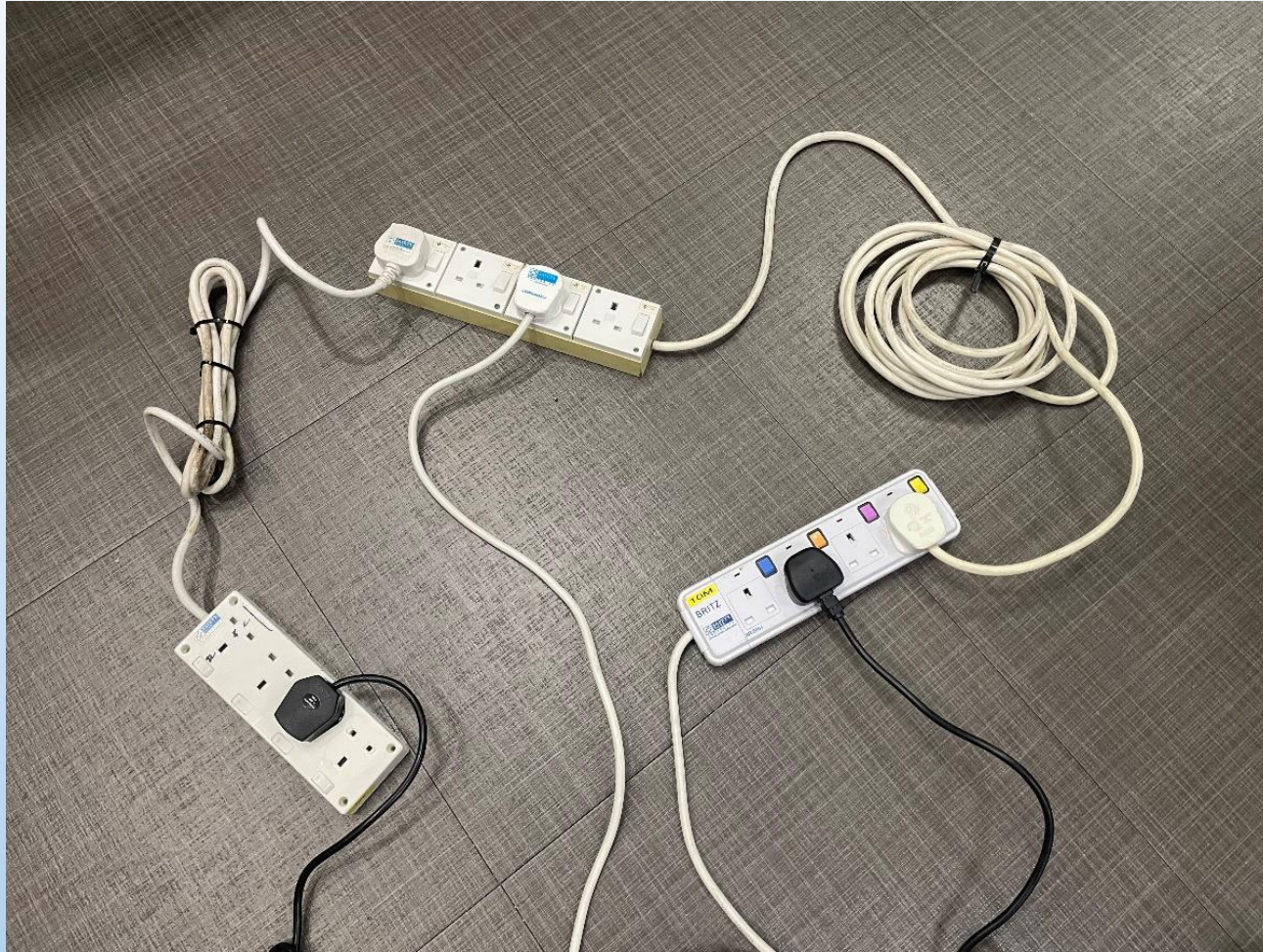
**Equipment that is plugged into the electrical socket for electrical power supply via the universal travel adapter shall not be left unattended.** The electrical power supply shall be switched off or disconnected if the equipment is to be left unattended.

Universal travel adapter shall conform to appropriate international safety standards such as IEC60884, or the European Low Voltage Directive 2006/95/EC. Products complying with the European Low Voltage Directive 2006/95/EC bear the "CE mark".

# Usage of Extension Cords

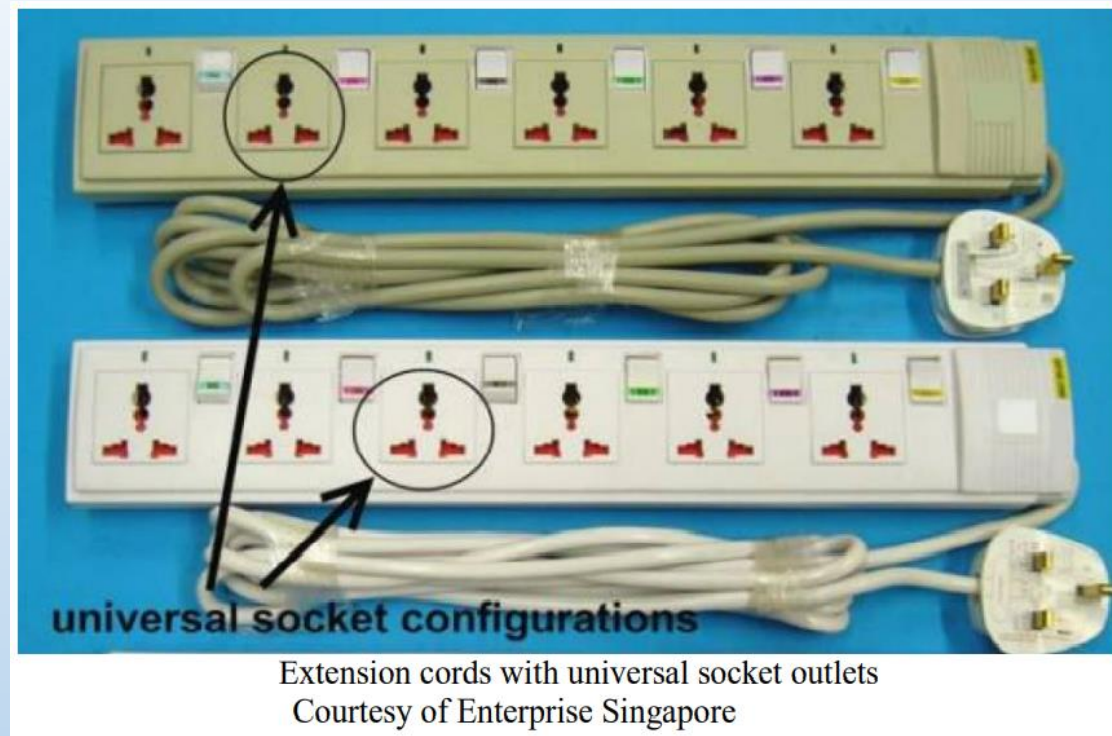


- All newly purchased extension cords used in NUS shall bear the “Safety Mark” certification or conform to appropriate international safety standards such as BS1363, IEC60884, or the European Directive 93/68/EEC.
- Not for ambient temperature having a peak value exceeding 40°C, or average value over 24 hours exceeding 35°C.
- Not for atmosphere subject to abnormal pollution by smoke, chemical fumes, rain, spray, prolonged periods of high humidity or other abnormal conditions.
- No daisy chain – chain of multiple extension cords or extension cord with multi-plug adapters.



**Daisy Chaining – strictly not allowed**

# Disapproved Extension Cords



- Extension cords with universal socket outlet as shown above are not approved for use in NUS..
- These extension cords have a high tendency to overheat and result in fire hazard.
- The temperatures of the Live and Neutral contacts are found to exceed safety limits when loaded to the rated current of the sockets in Singapore.

# Covered / Closed-toes shoes Policy

Examples of covered / closed-toes shoes



Examples of non-covered / sandals / non closed-toes shoes (not allowed)



Ref: [https://share.nus.edu.sg/corporate/procedures/safety\\_and\\_health/General-Lab-Safety-Manuals/NUS%20General%20Lab%20Safety%20Manual.pdf](https://share.nus.edu.sg/corporate/procedures/safety_and_health/General-Lab-Safety-Manuals/NUS%20General%20Lab%20Safety%20Manual.pdf)

# CDE Safety Inspection Demerit Point System

## IMPLEMENTATION

This demerit point system is designed to affect both the offender and his Principle Investigator (PI). Every offence will see the demerit points allocated to both the offender and his/her PI.

### For the offender

1st point: Issue verbal warning

2nd point onwards: Issue written warning

3rd point onwards:

1. Suspend the offender from using any laboratory for 2 weeks.
2. Offender to retake OSHE's relevant safety IVLE course.

### For the offender's PI

Points accumulated by the offender will also be recorded under the PI's research group or lab.

≥10 Points accumulated by all offenders: Written notification will be sent to the PI informing him/her of the offences

≥15 Points accumulated by all offenders: Written notification will be sent to the PI informing him/her of the additional offences AND a Safety timeout of 2 weeks will be imposed on all the PI's laboratories. A safety report on the corrective and preventive action have to be submitted before the timeout is lifted.

All points accumulated will be reset after 1 year.

# CDE Safety Inspection Demerit Point System

| S/N | Issue                                                                                                                                                                                                                                                    | Demerit Points |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1.  | <b>Housekeeping/General Safety</b>                                                                                                                                                                                                                       |                |
| a.  | Emergency exit doors and emergency passageways are visually or physically obstructed.                                                                                                                                                                    | 2              |
| b.  | Combustible materials (e.g. carton boxes and papers) in the work-space are not stored in safe manner. E.g.<br>a) Stored on flammable safety cabinets<br>b) Stored near ignition source<br>c) Accumulated below workbenches or stacked up to the ceiling. | 1              |
| c.  | There is slip, trip and fall hazard (e.g. wet floor, cable, wiring and tubing running across the walkways).                                                                                                                                              | 1              |
| d.  | Hazard communication was not evident and/or maintained in the workspace (e.g. laboratory warning sign, GHS label on chemical bottle, safety signage etc.)                                                                                                | 1              |
| e.  | Food and drink are evident in the workspace.                                                                                                                                                                                                             | 1              |
| f.  | Access to workspace is not controlled. Examples,<br>a) Laboratory door is intentionally propped open<br>b) Faulty access control                                                                                                                         | 1              |
| 2.  | <b>Personal Protective Equipment</b>                                                                                                                                                                                                                     |                |
| a.  | Eye protection requirement is not abided by personnel in workspace                                                                                                                                                                                       | 2              |
| b.  | Other PPE required for working in the work-space is not abided by personnel.                                                                                                                                                                             | 1              |
| 3.  | <b>Electrical Safety</b>                                                                                                                                                                                                                                 |                |
| a.  | There are unsafe electrical practices in workspace e.g. overloading of circuit, daisy-chaining, using of adapters without Safety Mark and leaving it unattended etc.                                                                                     | 1              |
| b.  | Electrical equipment (including cords & adaptors) used in the workspace are damaged                                                                                                                                                                      | 1              |
| c.  | Applicable electrical equipment used in the work-space did not come with "Safety Mark" label or equivalent.                                                                                                                                              | 1              |

|    |                                                                                                                                                                                                                                                                    |                |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 4. | <b>Radiation Safety (Ionizing and Non-ionizing)</b>                                                                                                                                                                                                                | Demerit Points |
| a. | A laser control area is not established for use of laser which exposure is above the Permissible Exposure Limits (PELs)                                                                                                                                            | 1              |
| 5. | <b>Local Exhaust Ventilation</b>                                                                                                                                                                                                                                   |                |
| a. | Activities using hazardous materials, which generate hazardous fumes and/or aerosols, are conducted without appropriate LEV (e.g. fume hood, biosafety cabinet, snorkel arm, glove box etc.)                                                                       | 2              |
| b. | LEV is not tested according to manufacturer recommendation                                                                                                                                                                                                         | 1              |
| 6. | <b>Chemical Safety</b>                                                                                                                                                                                                                                             |                |
| a. | Quantity for petroleum and flammable material has exceeded the Maximum allowable quantity (MAQ). Example,<br>a) MAQ per laboratory unit<br>b) Storage of more than 10% of flammable chemicals outside flammable safety cabinet                                     | 2              |
| b. | Regulated chemicals are not stored, handled and/or disposed accordance to regulatory requirements. Examples, Chemical weapons (CWC), Explosive precursor (EP) & Poison<br>a) Not under lock and key.<br>b) No inventory and/or tracking of usage, where applicable | 2              |
| c. | Flammable liquid and/or gas is stored in close proximity to oxidisers, heat and other ignition sources. (e.g. Electrical sockets, oven, furnaces, etc.)                                                                                                            | 2              |
| d. | Chemicals are not stored in accordance to their chemical compatibility.                                                                                                                                                                                            | 1              |
| e. | Secondary containment is used for safe storage of chemicals and hazardous waste, where necessary.                                                                                                                                                                  | 1              |
| f. | Gas cylinders are not secured or chained to prevent from toppling or falling.                                                                                                                                                                                      | 1              |

|     |                                                                                                                                                                                                             |   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 7.  | <b>Biosafety</b>                                                                                                                                                                                            |   |
| a.  | Autoclaves are not maintained & certified according to regulatory requirement                                                                                                                               | 2 |
| b.  | Sharps are not managed in a safe manner. e.g.<br>a) using non puncture-proof containers and not kept closed at all times or placed on the floor<br>b) over-filling of container<br>c) re-capping of needles | 1 |
| 8.  | <b>Change of Use</b>                                                                                                                                                                                        |   |
| a.  | Activities using hazardous chemicals are evident in the office area.                                                                                                                                        | 2 |
| 9.  | <b>Emergency Preparedness</b>                                                                                                                                                                               |   |
| a.  | Emergency equipment and/or protection/detection system are visually or physically obstructed.                                                                                                               | 2 |
| b.  | Emergency equipment (e.g. first aid box, emergency eye wash and shower, fire extinguisher) are not provided or maintained.                                                                                  | 1 |
| 10. | <b>Energy Conservation</b>                                                                                                                                                                                  |   |
| a.  | Fume hood sash is not lowered                                                                                                                                                                               | 1 |
| b.  | Equipment are not switched off when not in use                                                                                                                                                              | 1 |

# Use Storage Facilities provided



**E2A Design Studios**

**Approach respective facility  
ICs for booking/usage of  
storage facilities**



**iDP Assembly Workshop**

# Emergency Services and Contacts

- Campus Security /All emergencies – 6874 1616
- OSHE main contact line – 6516 1084
- Medical assistance service - 6776 1631 / 6516 2880
- University Counselling Service - 6516 7777 (24-hrs)
- Ambulance Service - 995

Should there be any accident, contact the campus Security and your immediate project supervisor.

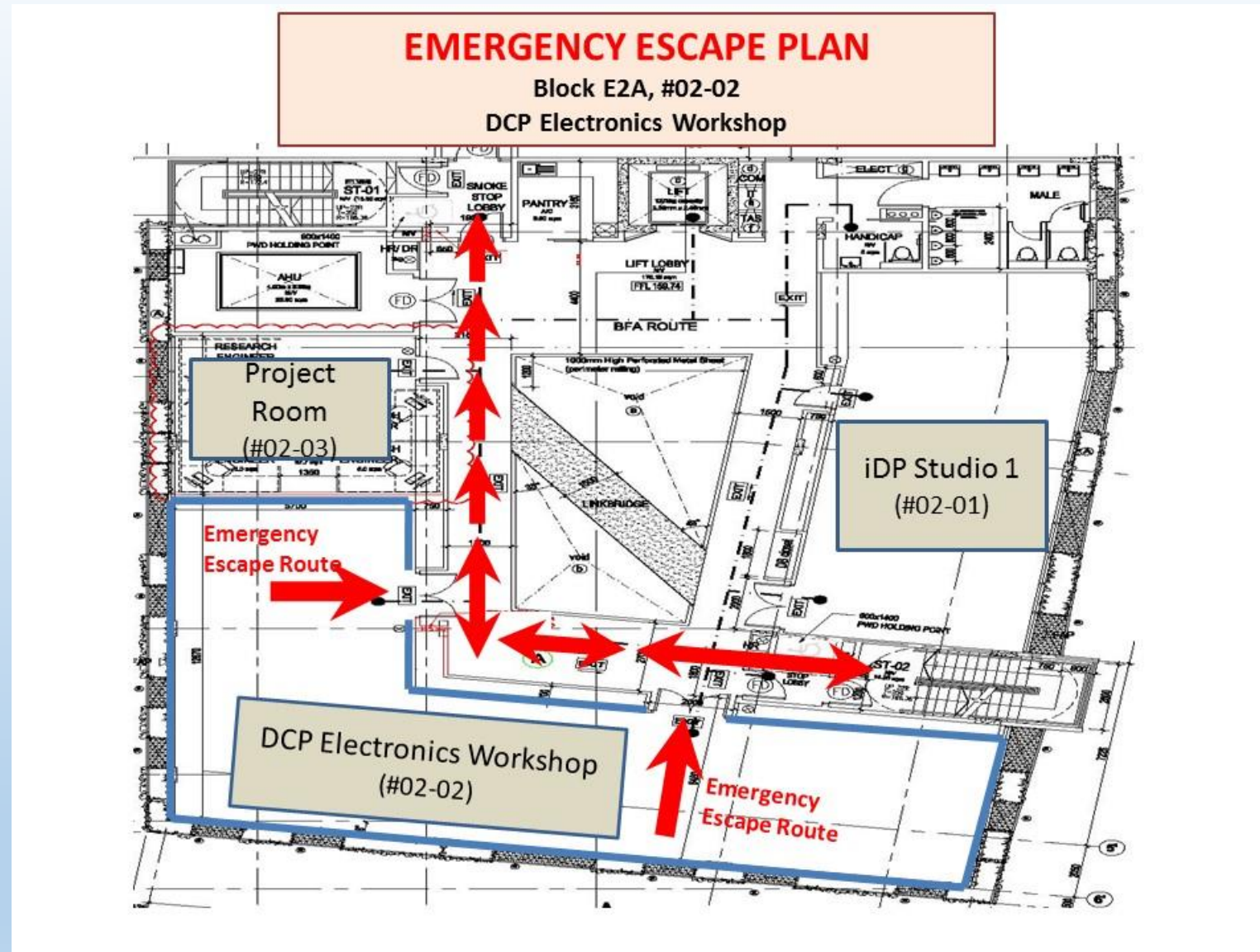
For minor injury like minor cuts on fingers, users shall make use of the First Aid box available in the lab for immediate treatment of the wound.

Further information on health and safety can be found on the website of the Office of Risk Management and Compliance (ORMC): <https://www.nus.edu.sg/ormc>

# In case of Fire Evacuation

- All personnel shall evacuate the building according to the Fire Warden / Safety Personnel
- Follow the Fire Evacuation Plan of the building and proceed to the designated Assembly Point
- Call Campus Security – 6874 1616

Further information on health and safety can be found on the website of the Office of Risk Management and Compliance (ORMC): <https://www.nus.edu.sg/ormc>



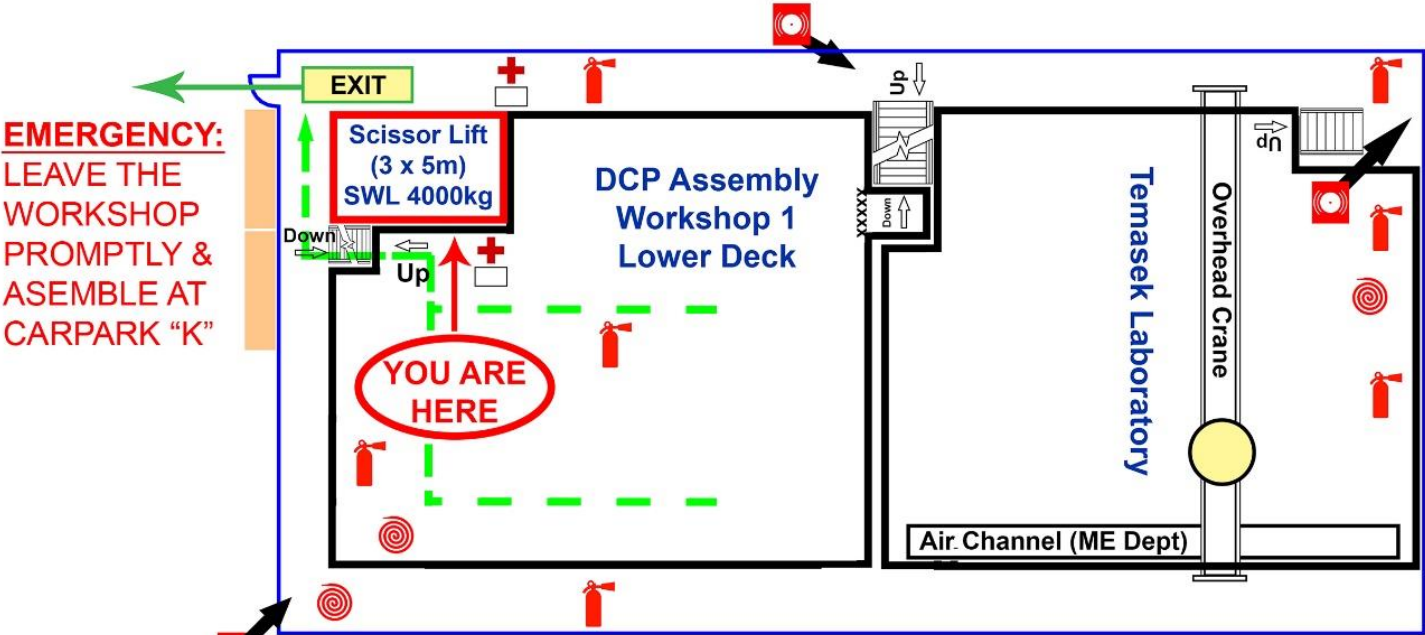
A copy of the Escape Plan can be found on the notice board of the lab

# Emergency Assembly Point for Blk E2A



A copy of the Assembly Point map can be found on the notice board of the lab

BLOCK E1A



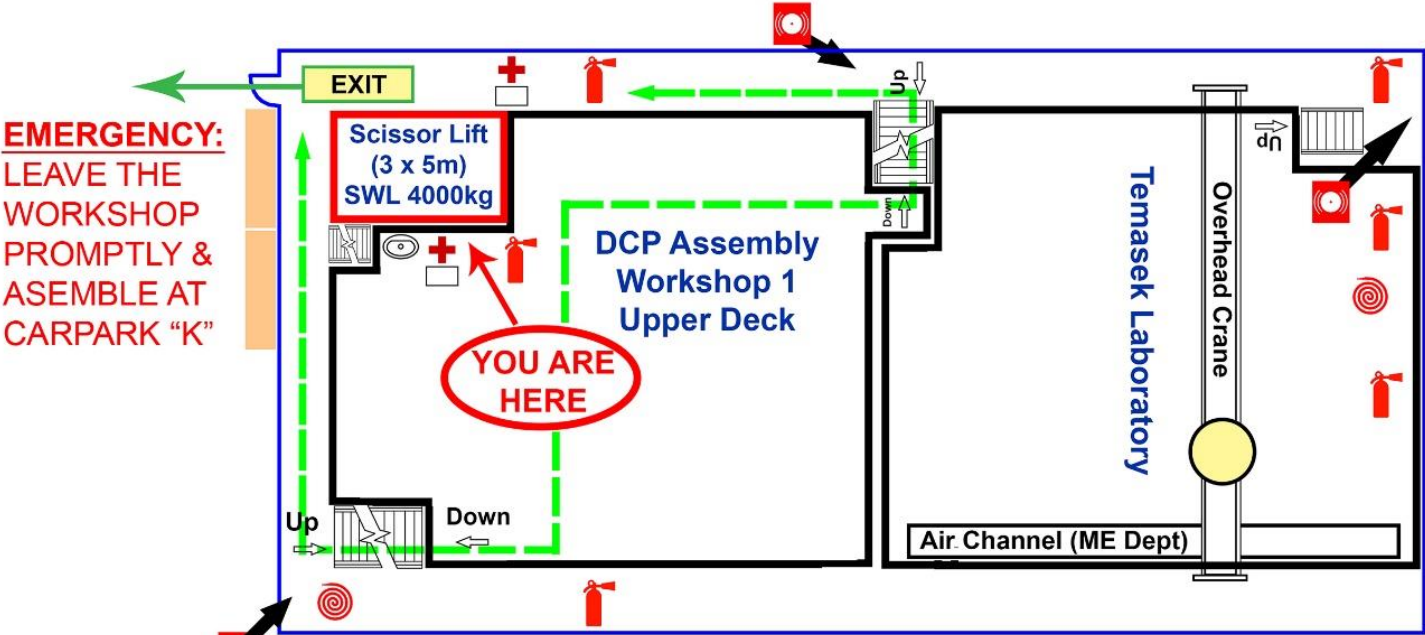
E1A-01-13 - LOWER DECK PLAN  
DCP ASSEMBLY WORKSHOP 1

|  |                   |  |                   |  |                   |  |                |
|--|-------------------|--|-------------------|--|-------------------|--|----------------|
|  | First Aid Box     |  | Fire Hose Reel    |  | Overhead Crane    |  | Roller Shutter |
|  | Fire Extinguisher |  | Fire Alarm Switch |  | Fire Escape Route |  | Locked Gate    |

NON - EMERGENCY CONTACT (Power failure, no water suply, etc) -- 6516 1515

iDP Assembly Workshop 1 (Lower Deck) Escape Plan

BLOCK E1A



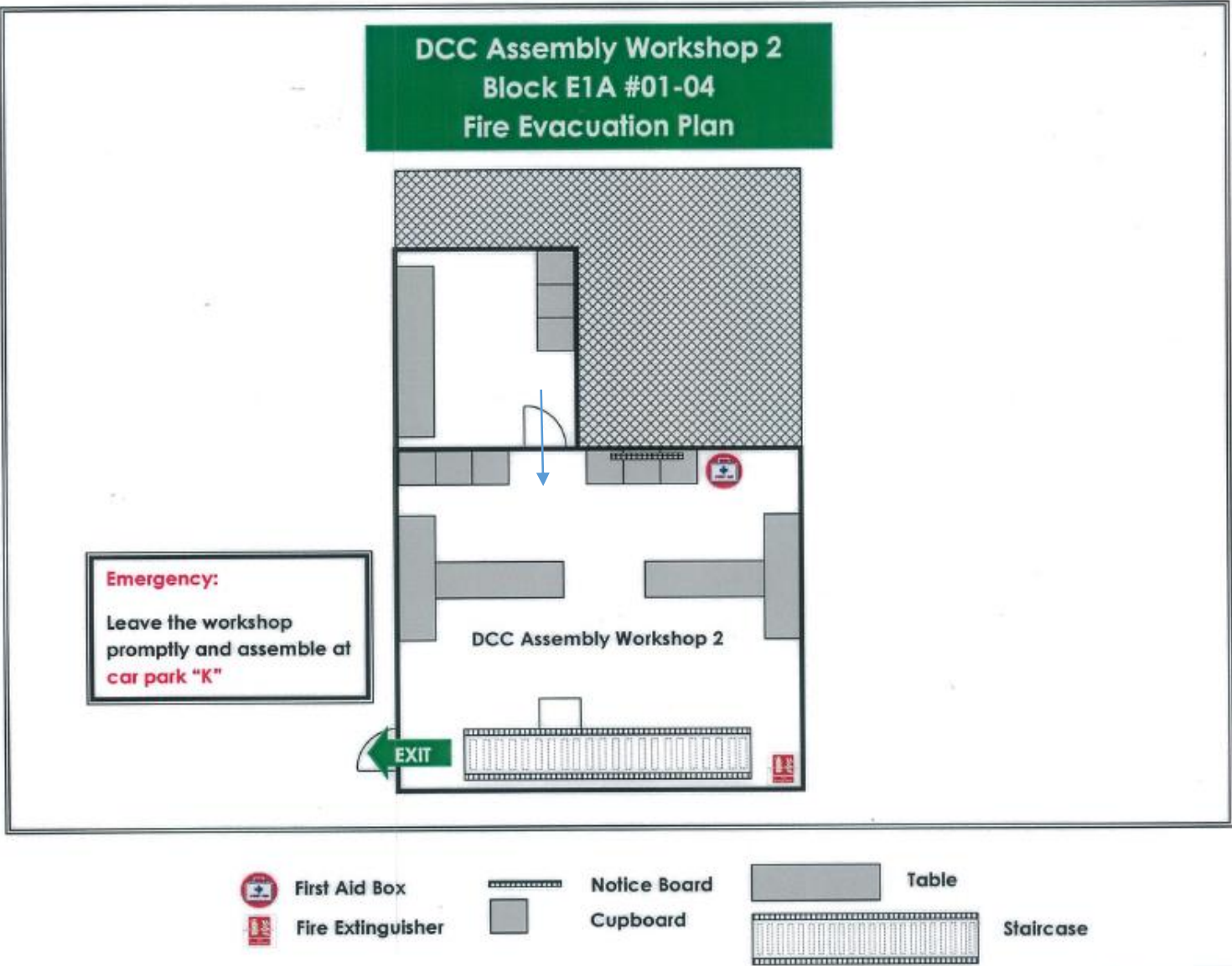
E1A-01-13 - UPPER DECK PLAN  
DCP ASSEMBLY WORKSHOP 1

|                                                                                     |                   |                                                                                       |                   |                                                                                       |                   |                                                                                       |                |
|-------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------|----------------|
|   | First Aid Box     |   | Fire Hose Reel    |   | Overhead Crane    |   | Roller Shutter |
|  | Fire Extinguisher |  | Fire Alarm Switch |  | Fire Escape Route |  | Water Basin    |

NON - EMERGENCY CONTACT (Power failure, no water suply, etc) -- 6516 1515

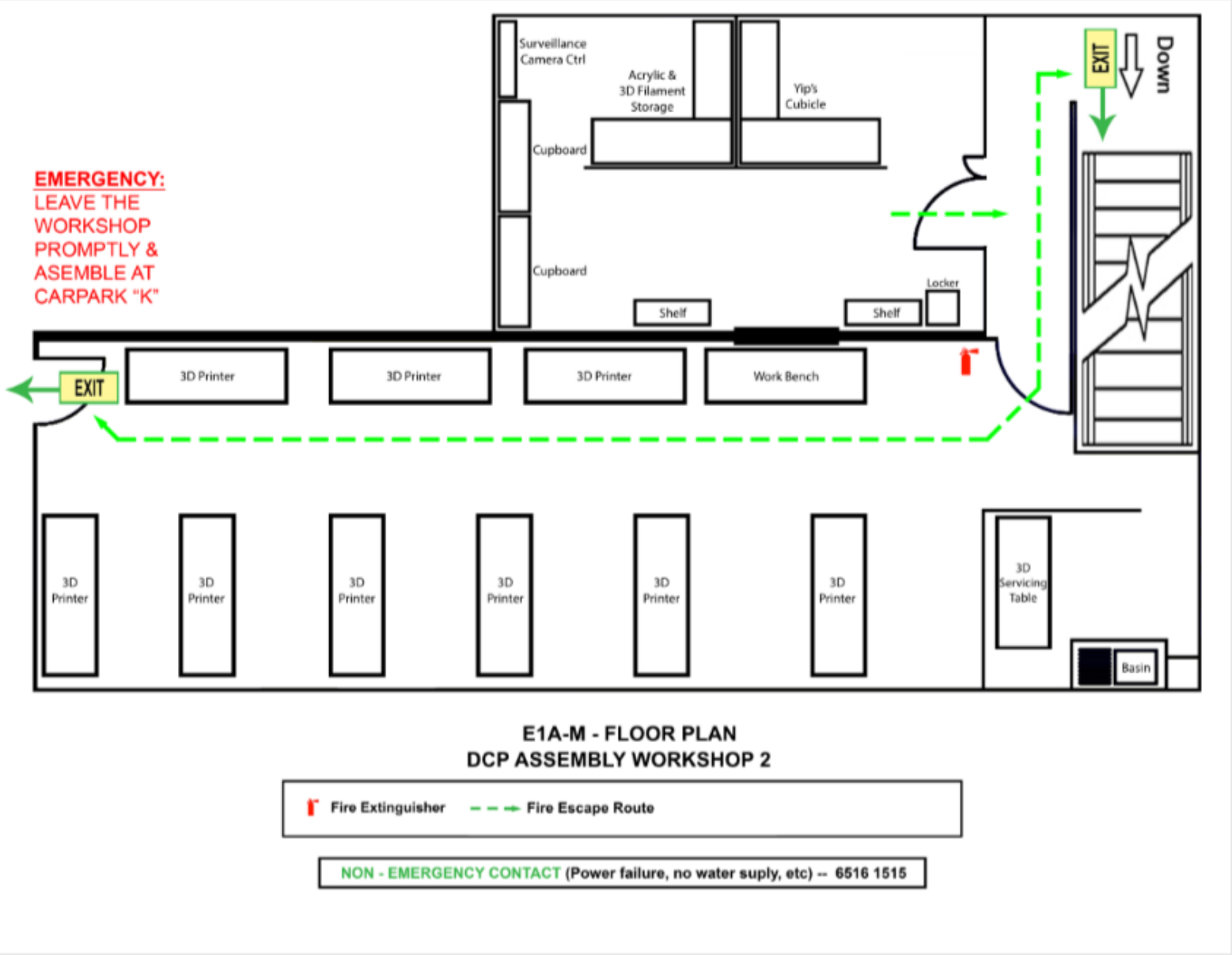
iDP Assembly Workshop 1 (Upper Deck) Escape Plan

BLOCK E1A



iDP Assembly Workshop 2 (Level 1) Escape Plan

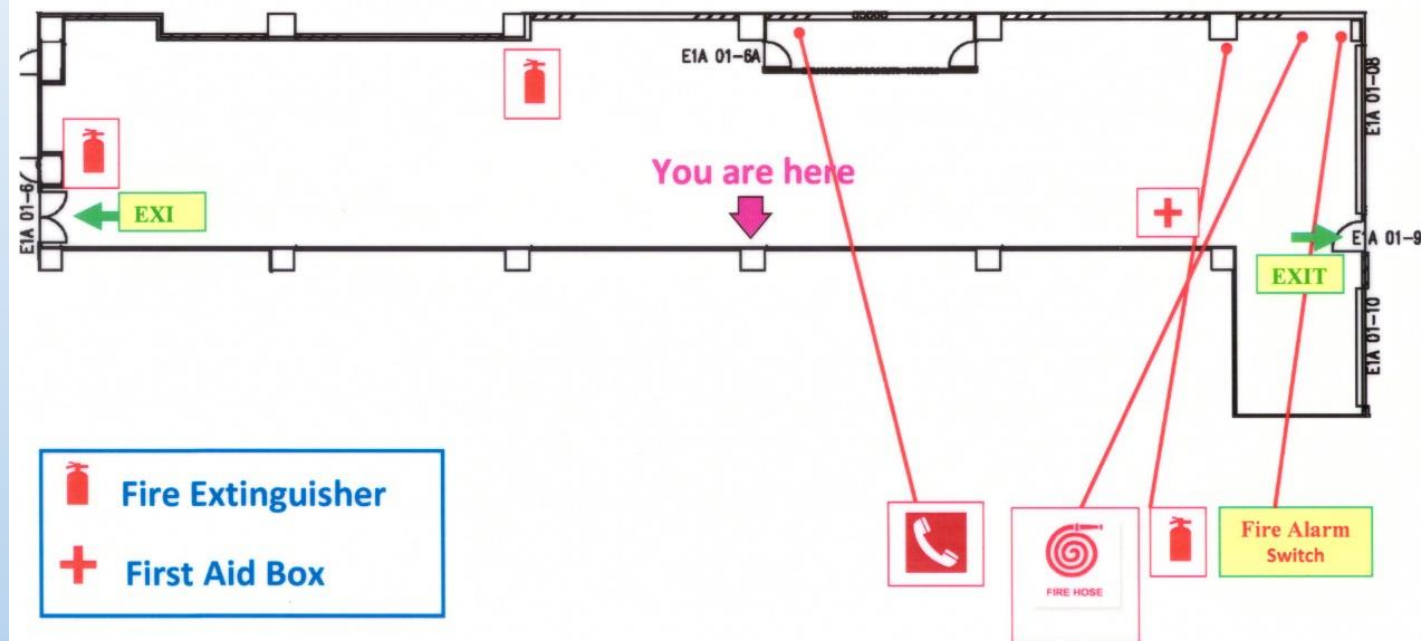
BLOCK E1A



iDP Assembly Workshop 2 (Mezzanine Level) Escape Plan

## BLOCK E1A

### E1A-01-06 Emergency Escape Route



Central Workshop - Training Workshop Escape Plan

# Emergency Assembly Point for Blk E1A

## Assembly Area

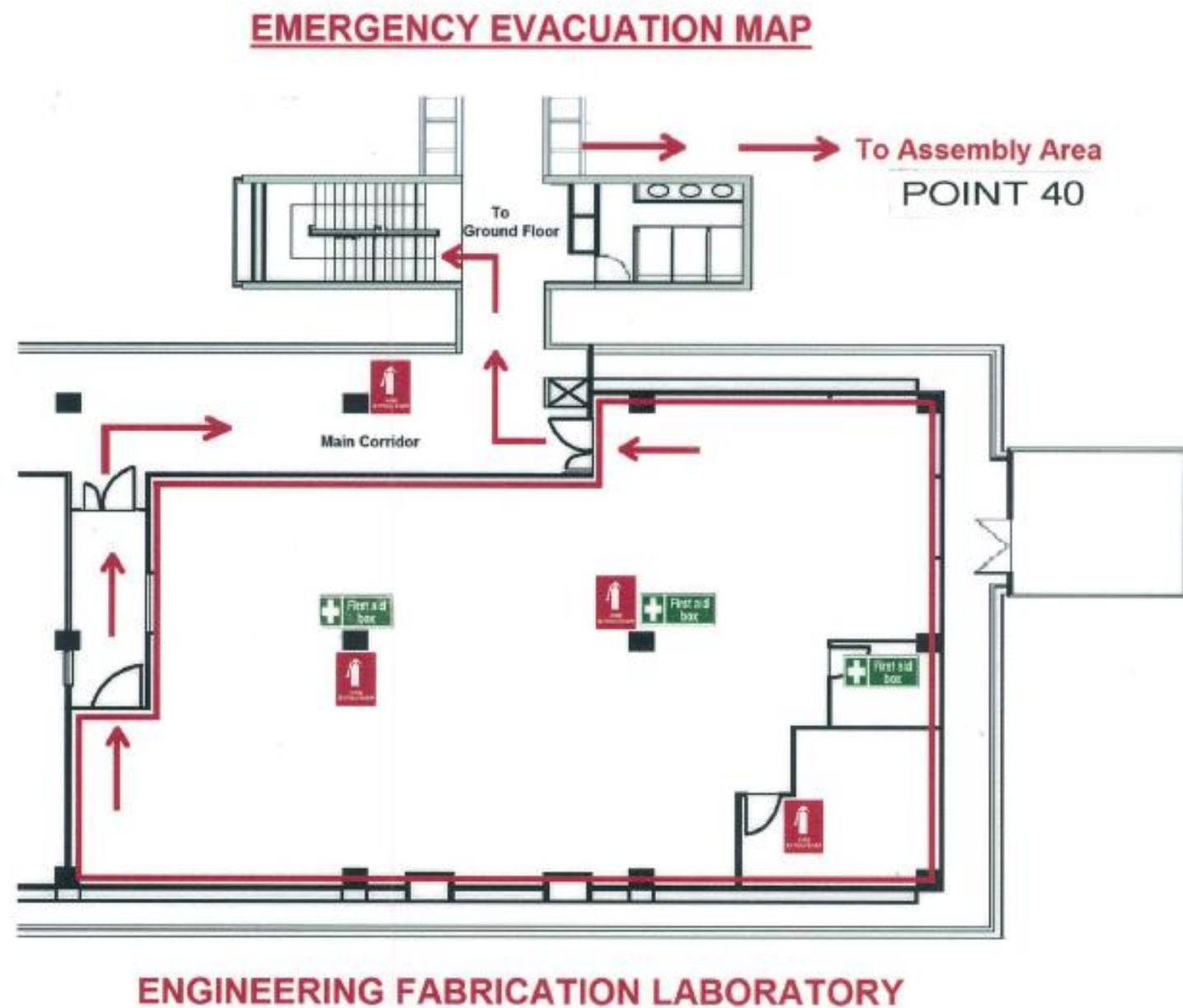
Location : Car Park “ K “

- E1,E1A , E2 ,EWS1,LT 1,LT 2 & LT 5



★ Proposed location of signage

## BLOCK E4



# Emergency Assembly Point for Blk E4

## Assembly Area

Location : Open area beside E4A

- BIK E4 & LT 6



★ Proposed location of signage

# Workshop / Lab User Registration

# SAMPLE OF WORKSHOP USER REGISTRATION FORM

Faculty of Engineering  
Design-Centric Curriculum

 **NUS**  
National University of Singapore

**USER REGISTRATION FOR WORKSHOP FACILITIES**

**WORKSHOP NAME / LOCATION:** DCC Electronics Workshop / E1A-03-12

| USER PARTICULARS                     |                   |
|--------------------------------------|-------------------|
| Name: _____                          | Matric No: _____  |
| Supervisor(s): _____                 |                   |
| Attachment Period: From: _____       | To: _____         |
| Contact No: _____                    | Email: _____      |
| <u>Emergency Contact Information</u> |                   |
| Name: _____                          | Contact No: _____ |

**PROJECT RISK ASSESSMENT**

Check the location

Fill in this part on your own

PS must be aware and Risk Evaluation must be submitted as evidence of safety briefing being conducted.

| SAFETY ISSUES                                     |                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------|
| Safety briefing conducted by Principal Supervisor | Yes [ <input type="checkbox"/> ], No [ <input type="checkbox"/> ] Date: _____ |

| SAFETY VIDEO VIEWING |                 |
|----------------------|-----------------|
| Safety Video Title   | Date of Viewing |
|                      |                 |

All mandatory videos to be viewed and date of viewing must be stated

Safety Quiz Score: \_\_\_\_\_

Fill up your Safety Quiz Score

# SAMPLE OF WORKSHOP USER REGISTRATION FORM

**PROJECT RISK ASSESSMENT**

1. To be completed by **Principal Supervisor**.  
2. Attach extra sheets if necessary.  
3. Applicable only to **DCC projects**; use FoE Risk Assessment Form according to the guidelines described by **OSHE/SOP/GL/05**.

Project Title: \_\_\_\_\_

**Risk Assessment:**

☐ No Risk

☐ Risk [Please tick the risks and indicate Assess Risks and Risks Control below]

☐ Mechanical    ☐ Electrical    ☐ Chemical    ☐ Biological    ☐ Noise

☐ Others (Please specify): \_\_\_\_\_

Assessed Risks: \_\_\_\_\_

Risk Controls: \_\_\_\_\_

Remark: \_\_\_\_\_

\_\_\_\_\_  
Name & Contact of Principal Supervisor

\_\_\_\_\_  
Signature and Date

Get your PS to fill up and sign

2<sup>nd</sup> Page, READ and sign off (remember to tick all the boxes)

**GUIDELINES FOR APPROPRIATE USE OF WORKSHOP FACILITIES**

1. Read the information on the **safety notice board** in the workshop and be aware of the **emergency actions and contact numbers**.
2. Report any incident or accident immediately to the **principal supervisor** during office hours and to the **campus security officer/faculty safety officer/principal supervisor** after office hours.
3. Do not work **ALONE** in the workshop at any time.
4. No operation of machines/equipment is allowed before any briefing on the respective safe operating procedure. Always follow the Standard Operating Procedures (SOPs).
5. Activities and usage of materials beyond those mentioned in the project risk assessment are **NOT** allowed.
6. Dispose waste (e.g. chemical and biological) in accordance to safety guidelines where appropriate.
7. Ensure all doors are properly locked if you are the last person leaving the workplace.
8. Wear proper attire in the workshop. **No sandals, slippers** or any **open-toed shoes** are allowed. Observe additional safety measures required at the nearby areas.
9. **Strictly NO FOOD is allowed in the workshop.**
10. Be responsible for your own safety and proper handling of your project setup. Think of the impact on others as a result of your action.
11. No Visitor is allowed in the workshop without permission from workshop supervisor or workshop administrator.
12. Sign the **attendance book** when accessing the workshop.
13. In violation of the above guidelines (rules and regulations), disciplinary actions including the suspension of privileges to the workshop will be enforced.

**DECLARATION**

☐ I agree to abide by the above Guidelines for Appropriate Use of Workshop Facilities.

☐ I shall not hold the University responsible or liable in the event of death, injury, disability or any mishaps while participating in the workshop activities.

☐ I have read, understood and hereby acknowledged receipt of an exact copy of this Registration form.

\_\_\_\_\_  
Name

\_\_\_\_\_  
Signature and Date

Ver 1.0 Updated on 20 July 2011

# Setting Expectations

## 1) House-keeping

- put rubbish into rubbish bins as soon as possible
- keep workplace clear of clutter and floor clean
- store project items in storage facilities when not in use
- do not eat in the workshop and labs as this will invite pests to the facilities
- drinking water should be deposited in the designated area
- return tools to the designated areas after usage

## 2) Time-management

- do not make last minute request to use facility/equipment
- take note that workshops officially close at 6PM

## 3) Duties and Responsibilities

- be cooperative with the workshop/lab staff
- user must bring along staff/student ID. Those without IDs may be refused entry.
- put on personal protective equipment (PPE) as required. It is a must to put on closed toes shoes in workshops.
- be alert of activities and happenings around you
- report any injury medical condition immediately ([https://inetapps.nus.edu.sg/osh/portal/eServices/ehs360\\_aims.html](https://inetapps.nus.edu.sg/osh/portal/eServices/ehs360_aims.html))
- report any facility/equipment/PPE fault to the Lab staff immediately
- know where the emergency switches, fire extinguishers and first aid boxes are
- always work in pairs in the workshops/labs
- always ask if in doubt.

# Safety Awareness Quiz

# **iDP Safety Awareness Quiz**

Please attempt the iDP Safety Awareness Quiz on the NUS Canvas (Safe845 – Quizzes)

In order to get registration to iDP Labs and Workshops, you need to score at least 17 out of 20 for the quiz.

You have unlimited attempts to pass the quiz within the stipulated period below.

You may start attempting the quiz from date/time to be confirmed.

The deadline for completing the quiz is date/time to be confirmed.